

CSE190/291

How The Web Tracks You / Targeted Advertising

Stateful Tracking, Part 2

UC San Diego

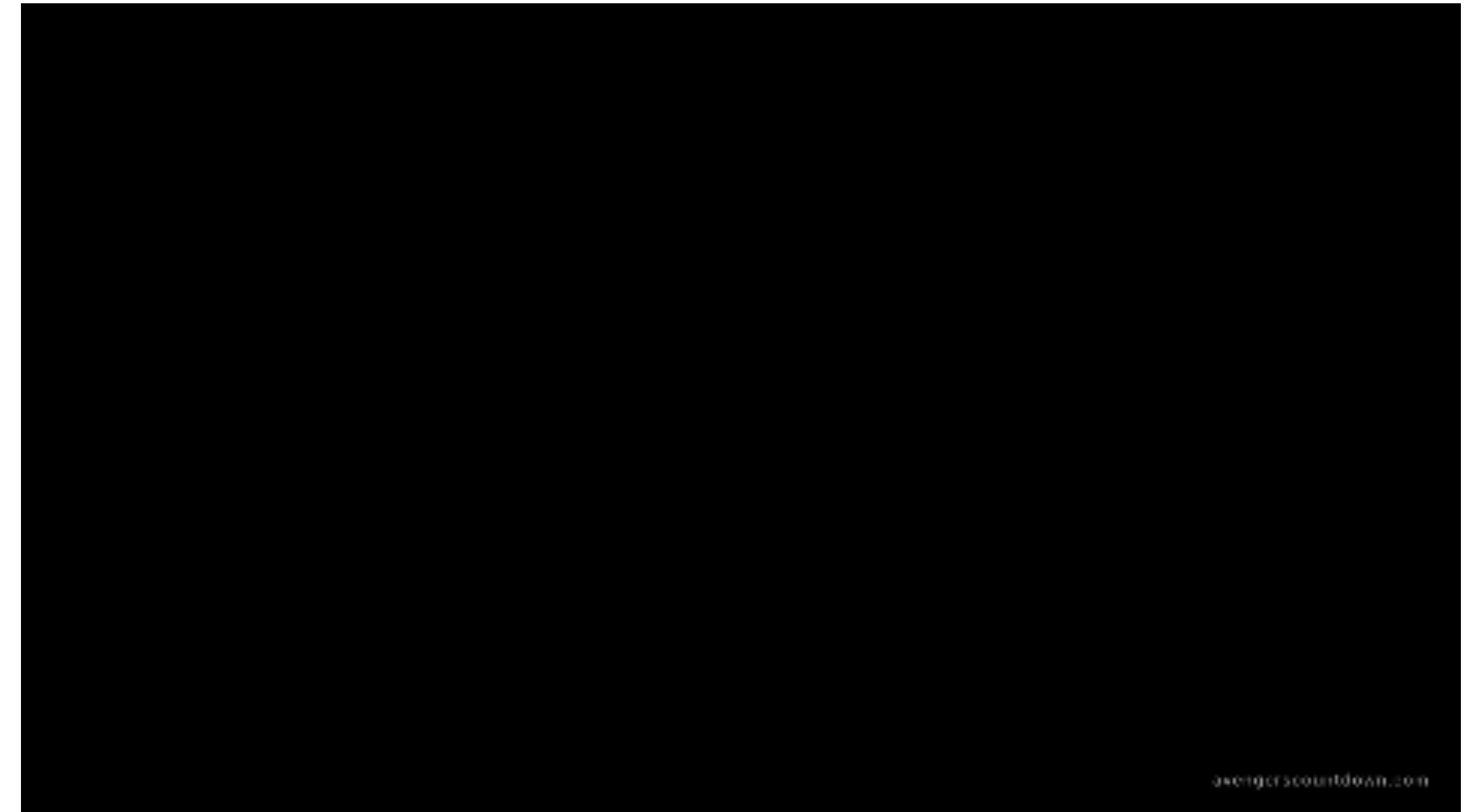
Housekeeping

General course things to know

- PA1 part 1 grades released, yay!
- PA1, parts 1 + 2 are released!
 - Part 2 due Tuesday, 10/13 (one week from today!)
- Implementing a full-scale “web application” that does some stateful tracking, using cookies, to track your behavior on a Google search competitor (Boogle)
 - Cookie-based tracking, databases, and server side logic
- **No class on 10/13!**
 - work on your PAs :)
- PA1, parts 3 + 4 *should* release on **10/14**

Previously on CSE190/CSE291...

- We talked about the details of web pages, servers, clients, and the construction of web pages (resources, origins, first-party / third-party, oh my)
- We began our discussion about cookies, how they work, and how they're scoped



Let's do a recap.

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 - Client (browser) sends them automatically with every subsequent request

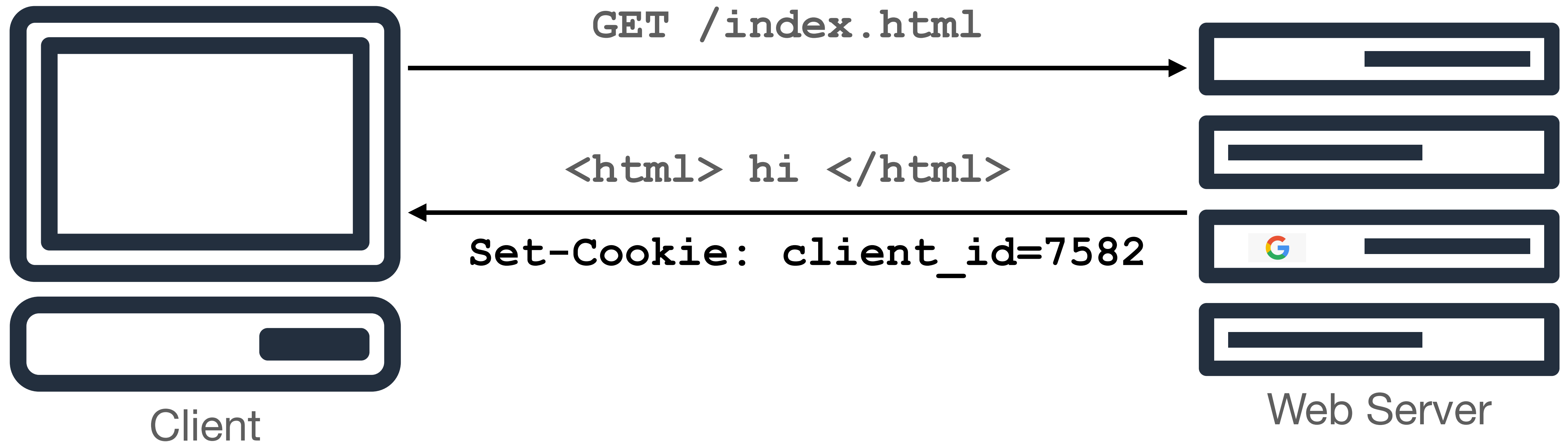
How do web cookies work?



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How do web cookies work?



Client

The web client
(browser) stores the
cookie value in a
local file.

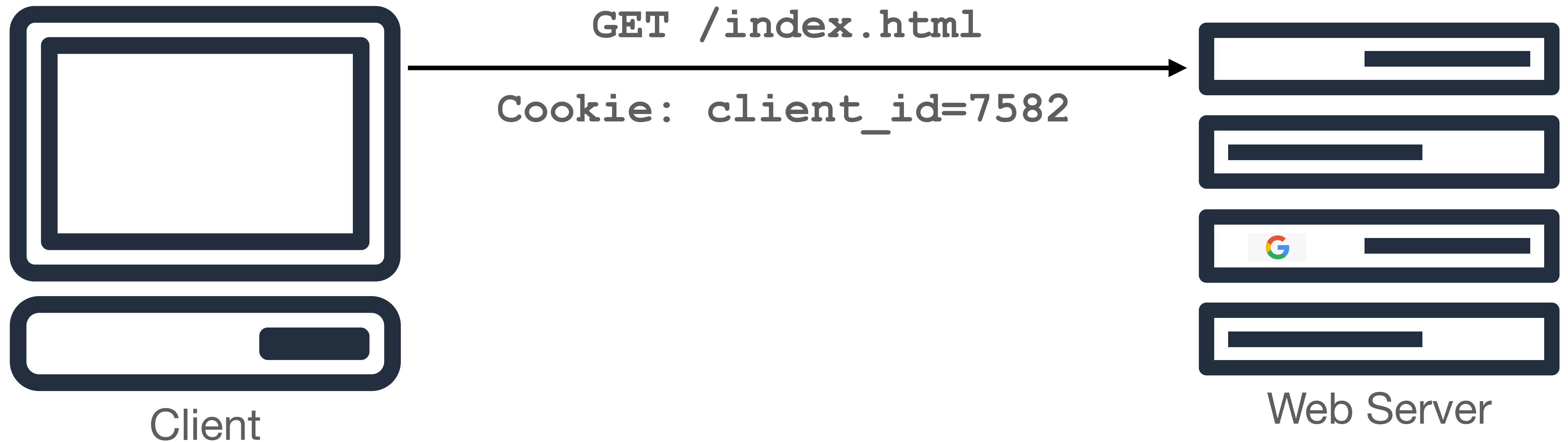


Web Server

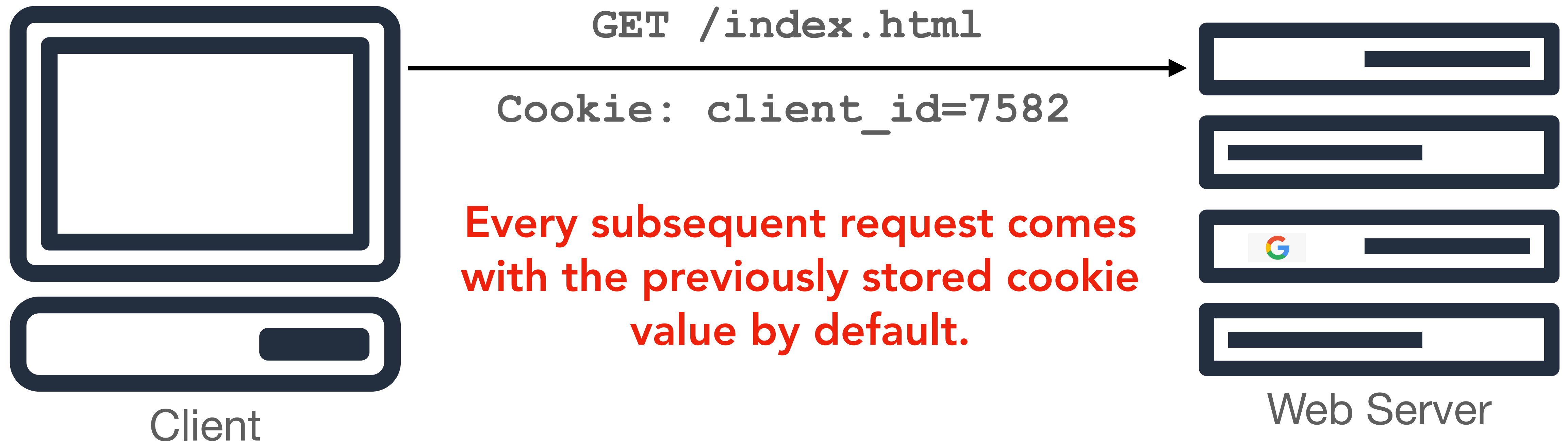
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- How do cookies get set and passed around?
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 - Client (browser) sends them automatically with every subsequent request
- Does the browser automatically send cookies to *any* server without checking anything?
 - No! Cookies have scopes!

Cookie Scoping

- How cookies are shared are ultimately **scoped** by the server, by several properties:
 - Domain — Set-Cookie: `client_id=7582; Domain=example.com`
 - Path — Set-Cookie: `client_id=7582; Path=/docs`
 - Secure — Set-Cookie: `client_id=7582; Secure`
 - How they're accessed — Set-Cookie: `client_id=7582; HttpOnly`
 - Time — Set-Cookie: `client_id=7582; Max-Age=3600`

Lots of other properties...

- `SameSite` (primarily a defense against a CSRF attack — see CSE 127)
- `Partitioned` (we'll talk about this in depth later)
- Thought experiment: **when a server gets a request with a cookie, what does it see?**

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- Cookie prefixes to the rescue!
 - E.g., Prepending your cookie name with `__Host-` tells the browser to do something special, and the server knows its original intention

How do we decide to send cookies? (Exam style)

Cookie 1:

name = mycookie

value = value

domain = login.site.com

path = /

Cookie 2:

name = mycookie2

value = value

domain = site.com

path = /

Cookie 3:

name = mycookie3

value = value

domain = site.com

path = /my/home

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Request to URL:	Cookie 1	Cookie 2	Cookie 3
<u>checkout.site.com</u>			
<u>login.site.com</u>			
<u>login.site.com/my/home</u>			
<u>site.com/my</u>			

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<u>login.site.com/my/home</u>	Yes	Yes	Yes
<u>site.com/my</u>	No	Yes	No

Today's lecture — Stateful Tracking, Pt. 2

Learning Objectives

- Begin to understand the stateful, third-party tracking ecosystem, the underlying cookies that support it, and what we might do about it
- Understand the fight and history behind third-party tracking and the fights against the fights against third-party tracking
- Learn about how, despite our best efforts to block cookies, cookies remain passed around the web

Stateful Tracking

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- With a group, answer the following questions:
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 - What you do on the page (e.g., make searches)
 - Where you visit, etc.

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 - Time spent on a page
 - Time spent on specific parts of a page
 - What you do on the page (e.g., make searches)
 - Where you visit, etc.
- **Goal of most trackers: To learn about your web history (where you’ve been) and use that to build a profile about you over time.**

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- *“The site that you intend to visit tracks things about you.”*
- Is this a privacy violation?

Defining First-Party Tracking

- *“The site that you intend to visit tracks things about you.”*
- Is this a privacy violation?
 - Not always — when you visit a website, you are “consenting” (in some implicit way) for the website to do whatever it wants with your interactions
 - Most browsers don’t do anything about first-party tracking (e.g., cookies), because, well, that’s just the Internet
- First-party tracking can still get companies into trouble though...

First-Party Problems

TECH

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

By [Kashmir Hill](#), Former Staff. Welcome to The Not-So Private Parts where technology & privacy collide

Published Feb 16, 2012, 11:02am EST, Updated Aug 11, 2022, 04:17am EDT

Recommendations can be creepy

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Companies might sell your data

July 15, 2026 | Press Release

Attorney General Paxton Secures \$150 Million Settlement Against 23andMe Over Data Breach That Exposed Genetic Information of 6.9 Million People

Companies might get breached
(and then go bankrupt)

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- With a group, answer the following questions:
 - What is web tracking and how do you define it?
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 - **What is third-party tracking?**

Third-Party Tracking

- How does a company actually *track* you on the web using cookies? Seems like there's a lot of protections?

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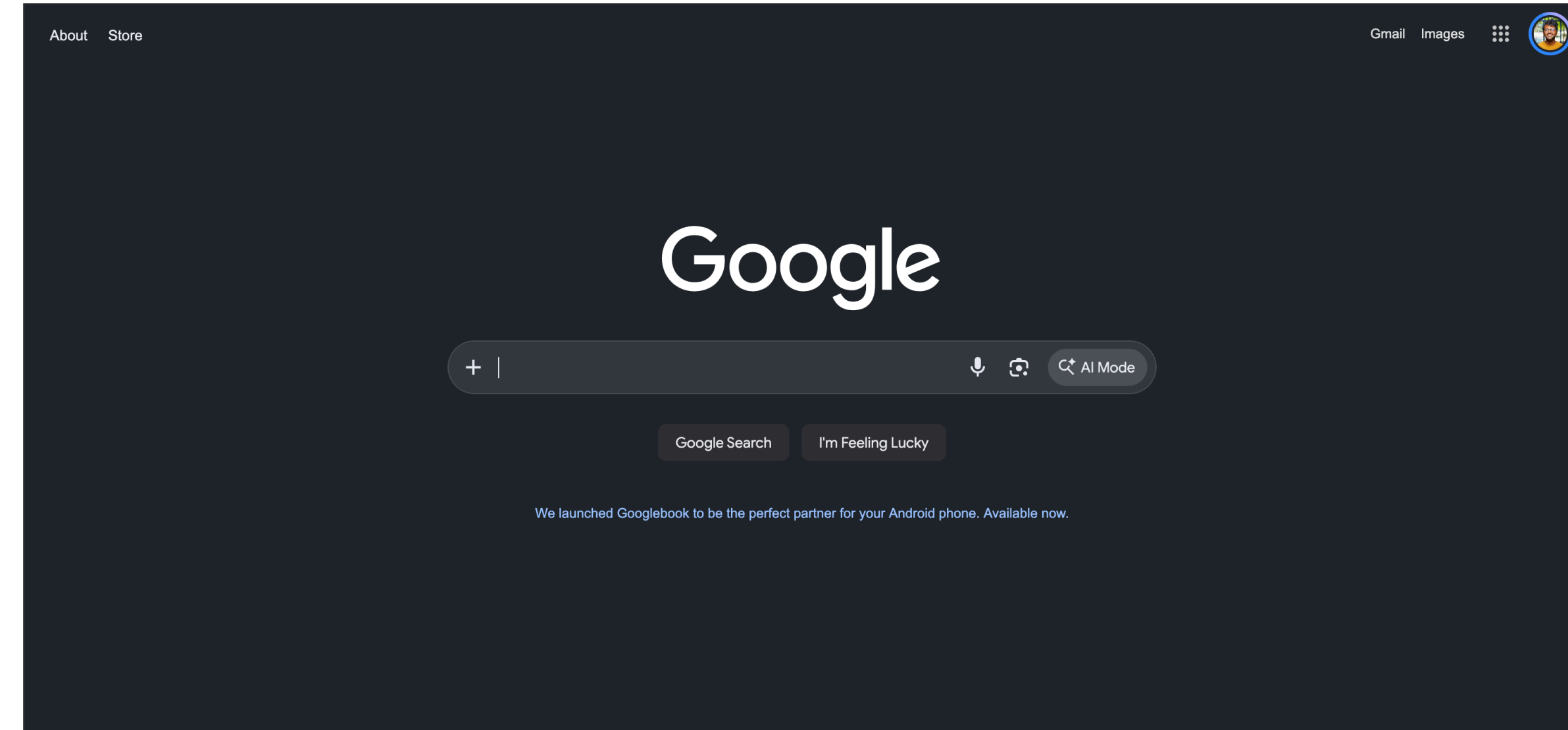
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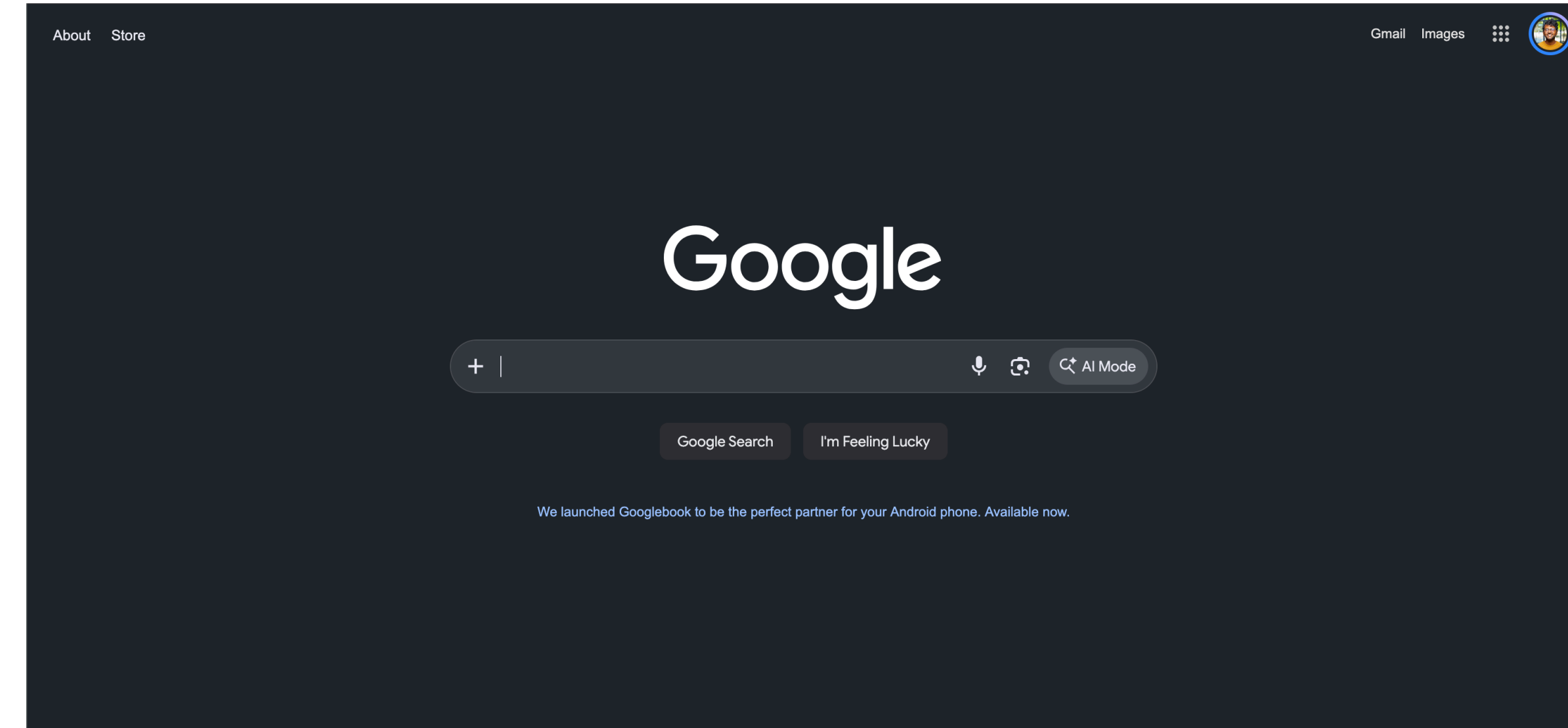
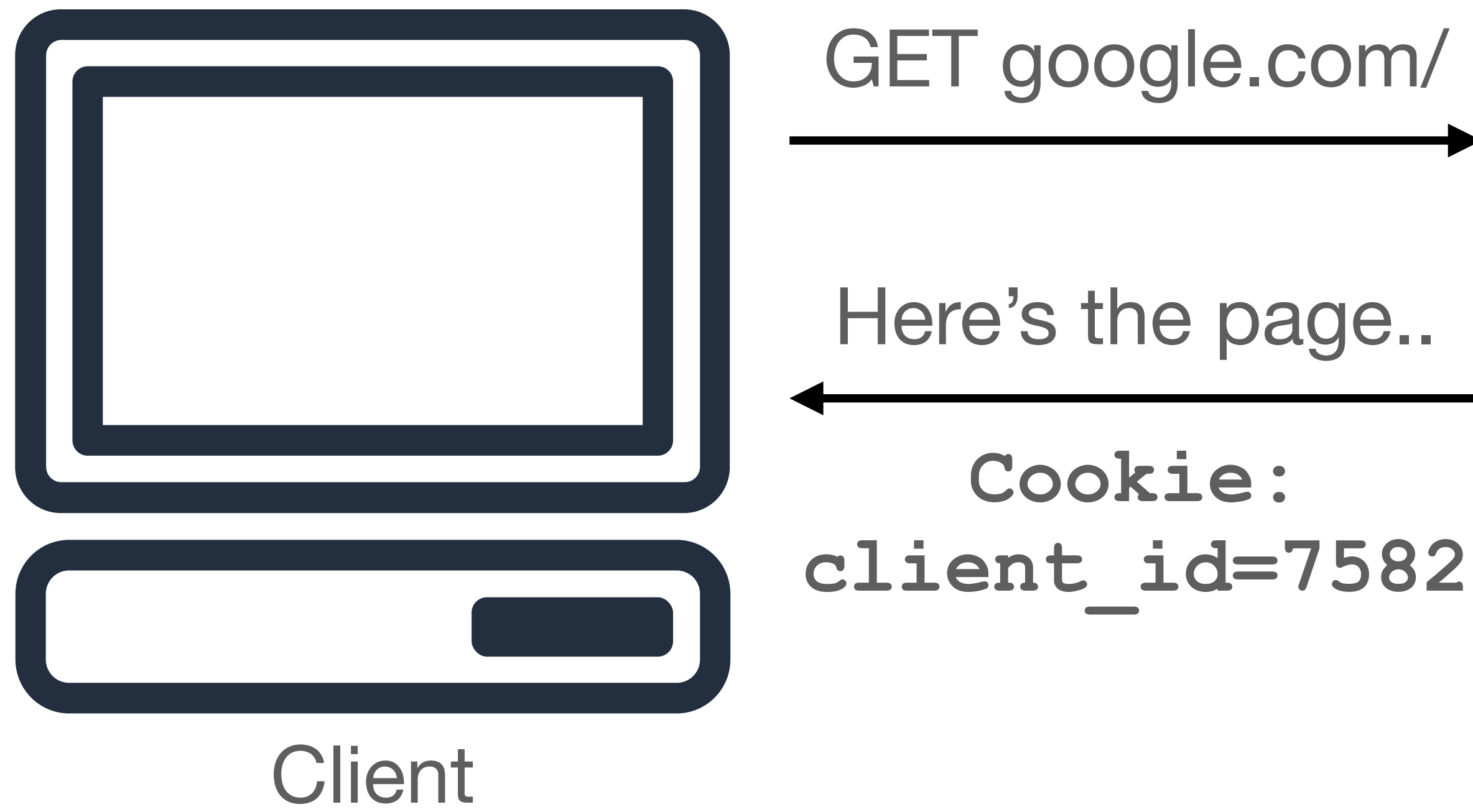
GET google.com/



google.com

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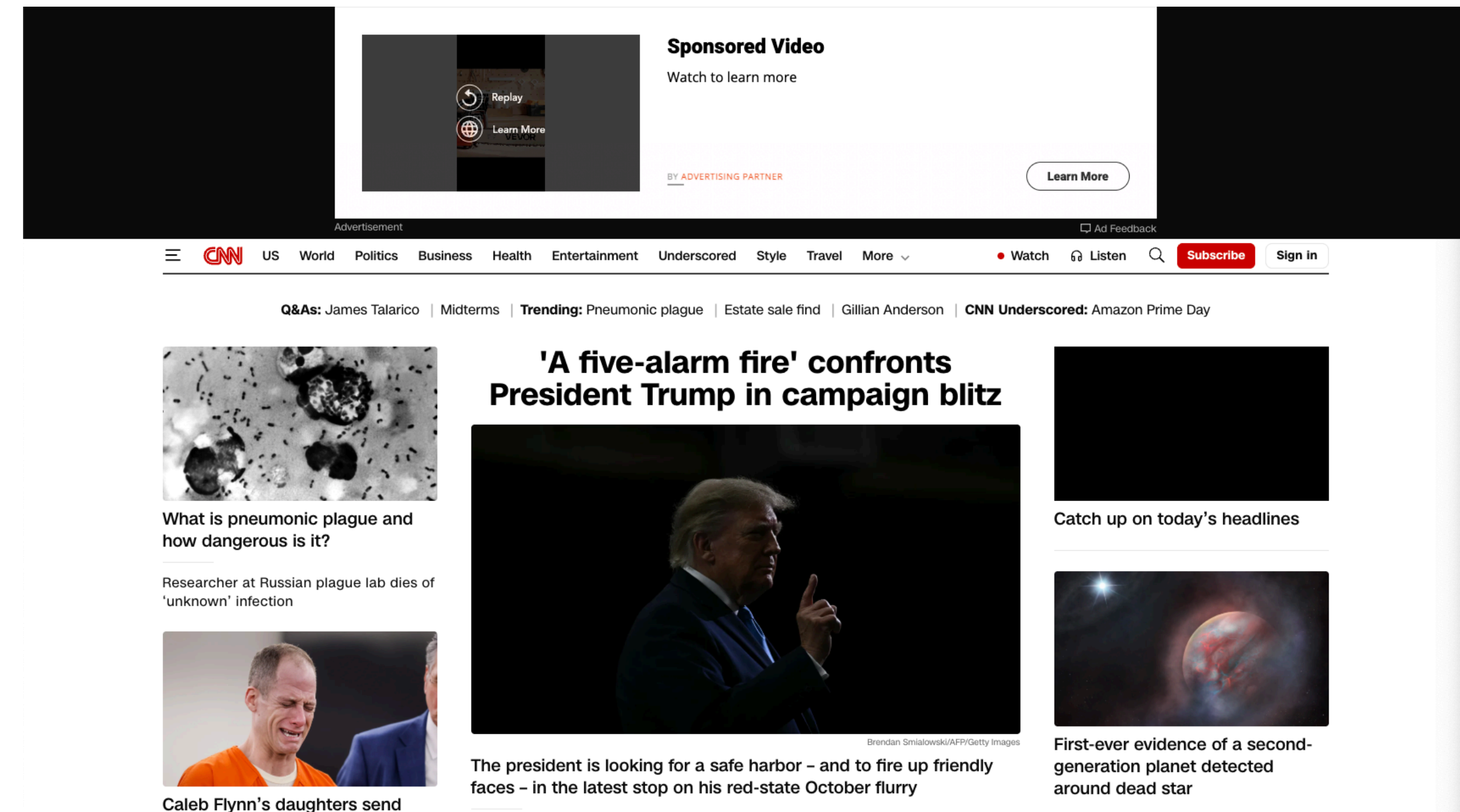
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GET cnn.com/



cnn.com

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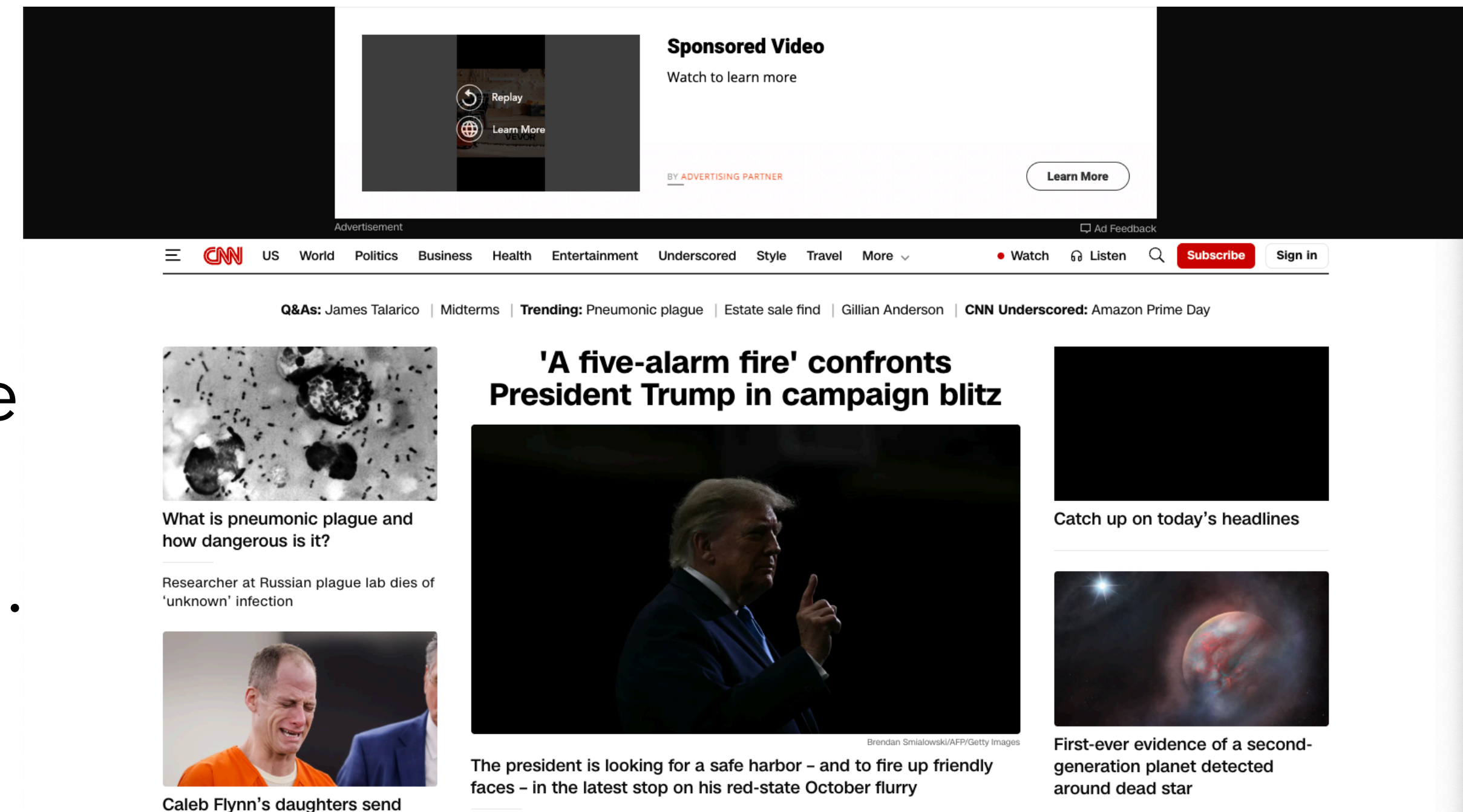
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Client

GET cnn.com/

By the time this page
has loaded, I have
already been tracked.
How?



cnn.com

Third-Party Tracking

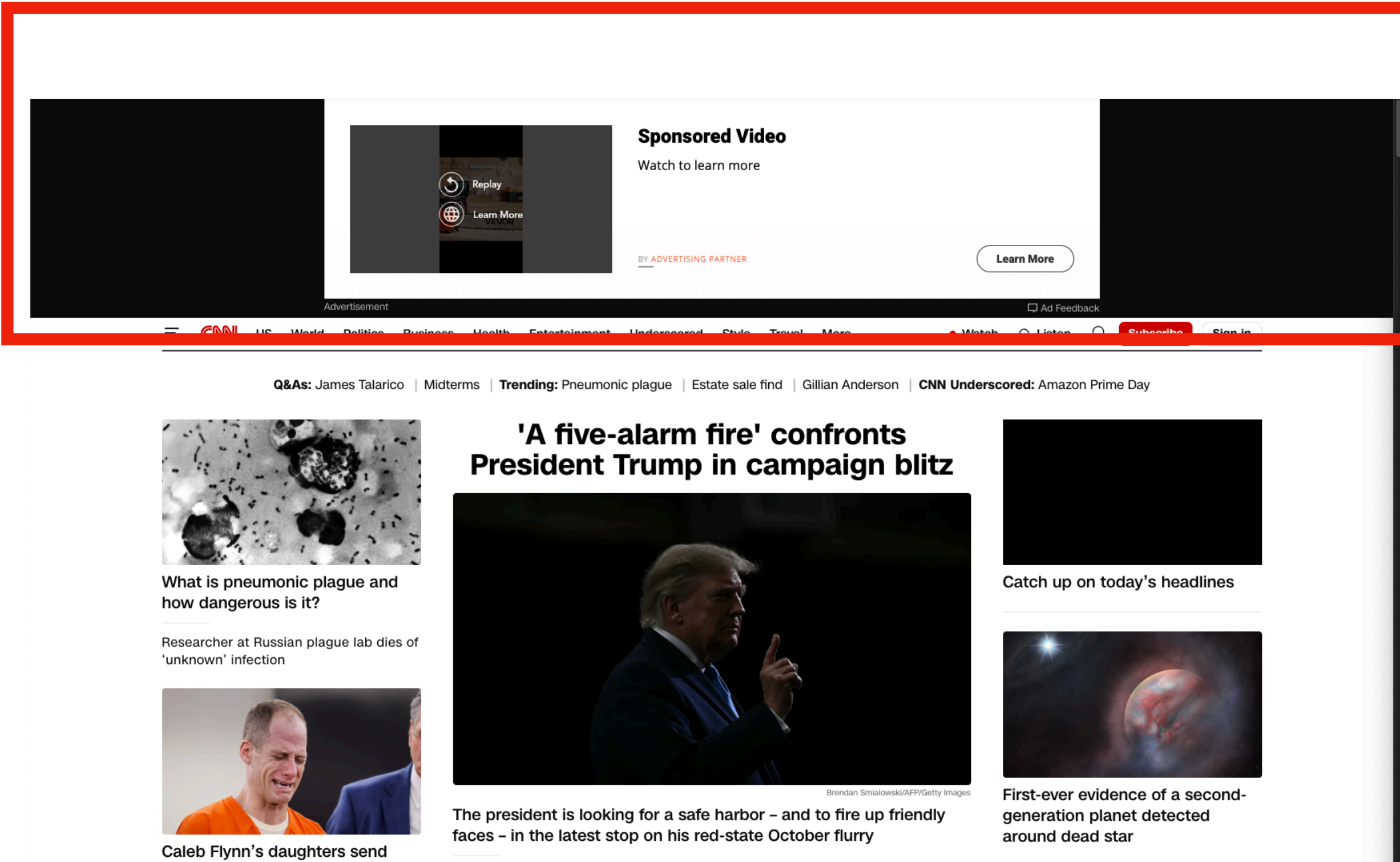
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Where did this ad load from?



Client

GET cnn.com/



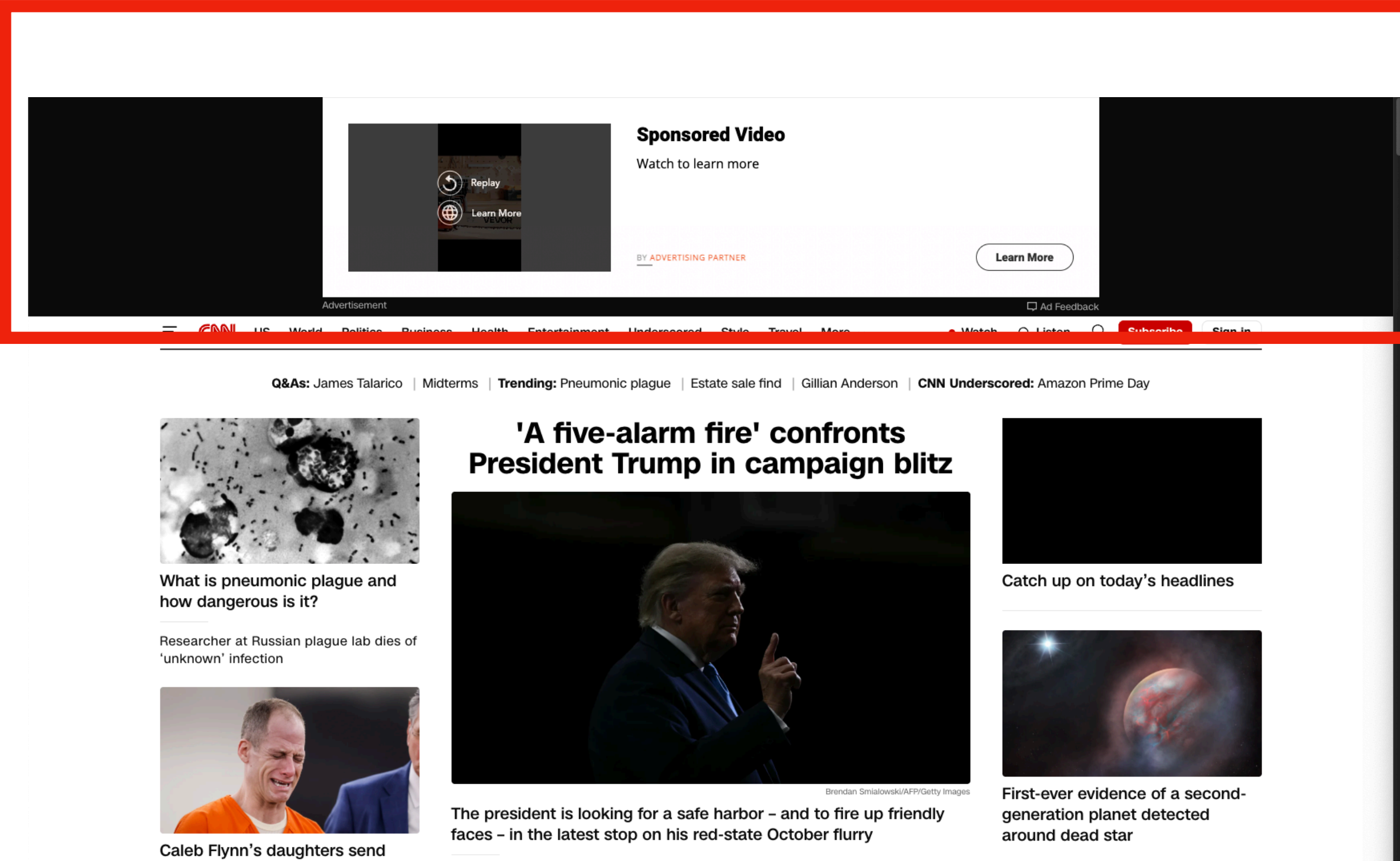
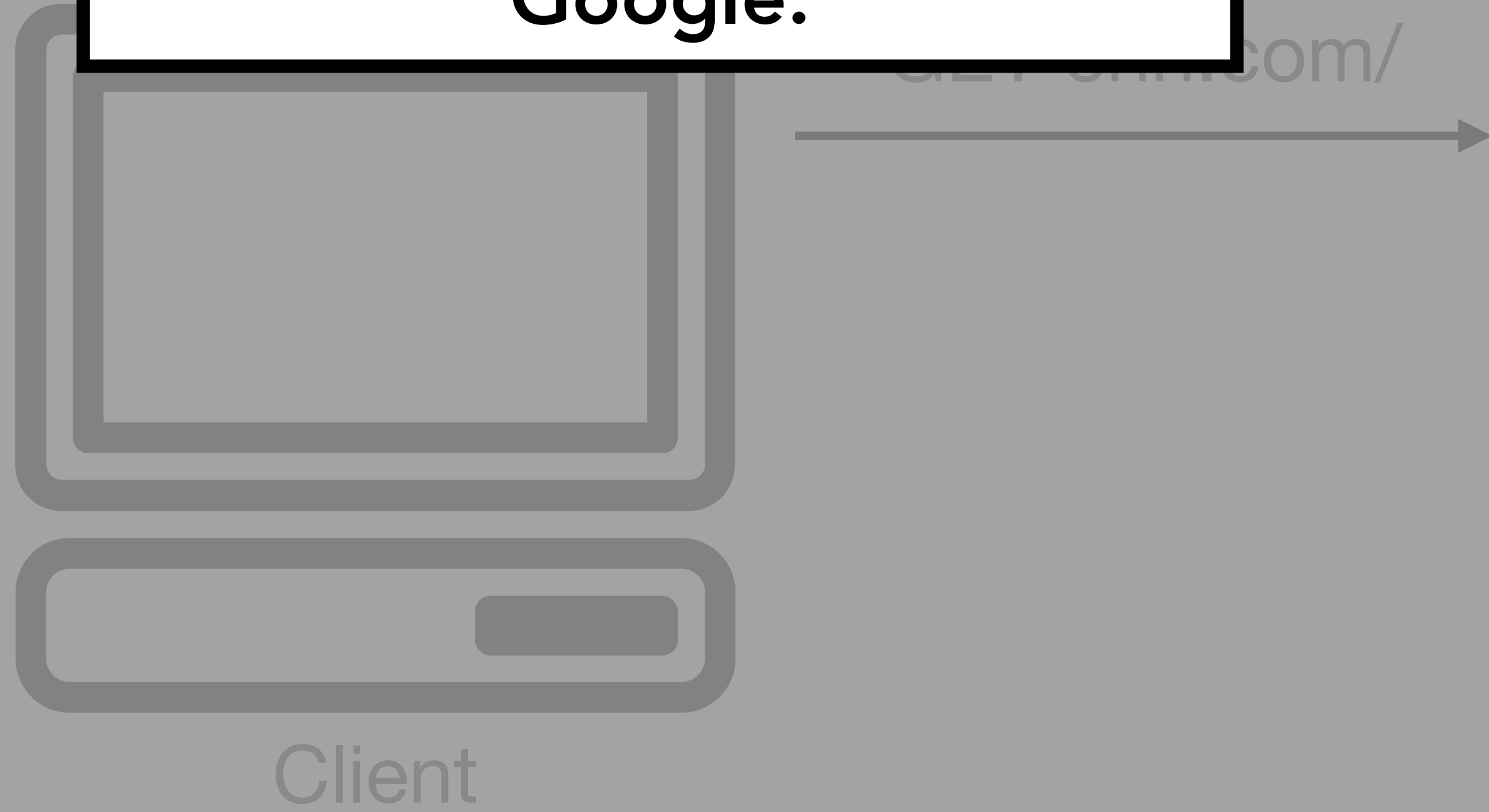
cnn.com

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Where did this ad load from?

Google.



cnn.com

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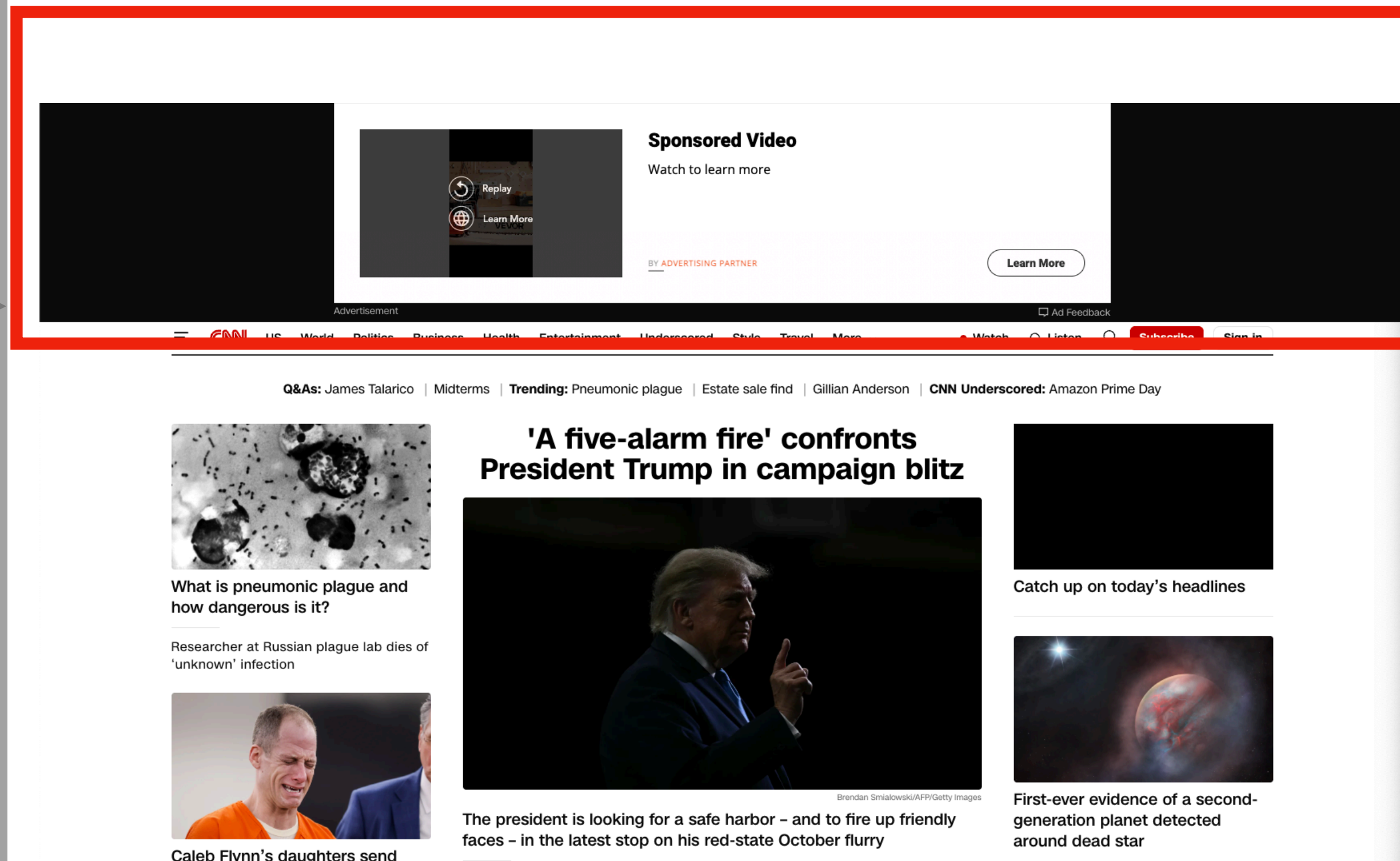
Where did this ad load from?

Google.

And what does the browser send when making requests to Google?

Client

on the web using cookies? Seems like



cnn.com

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Where did this ad load from?

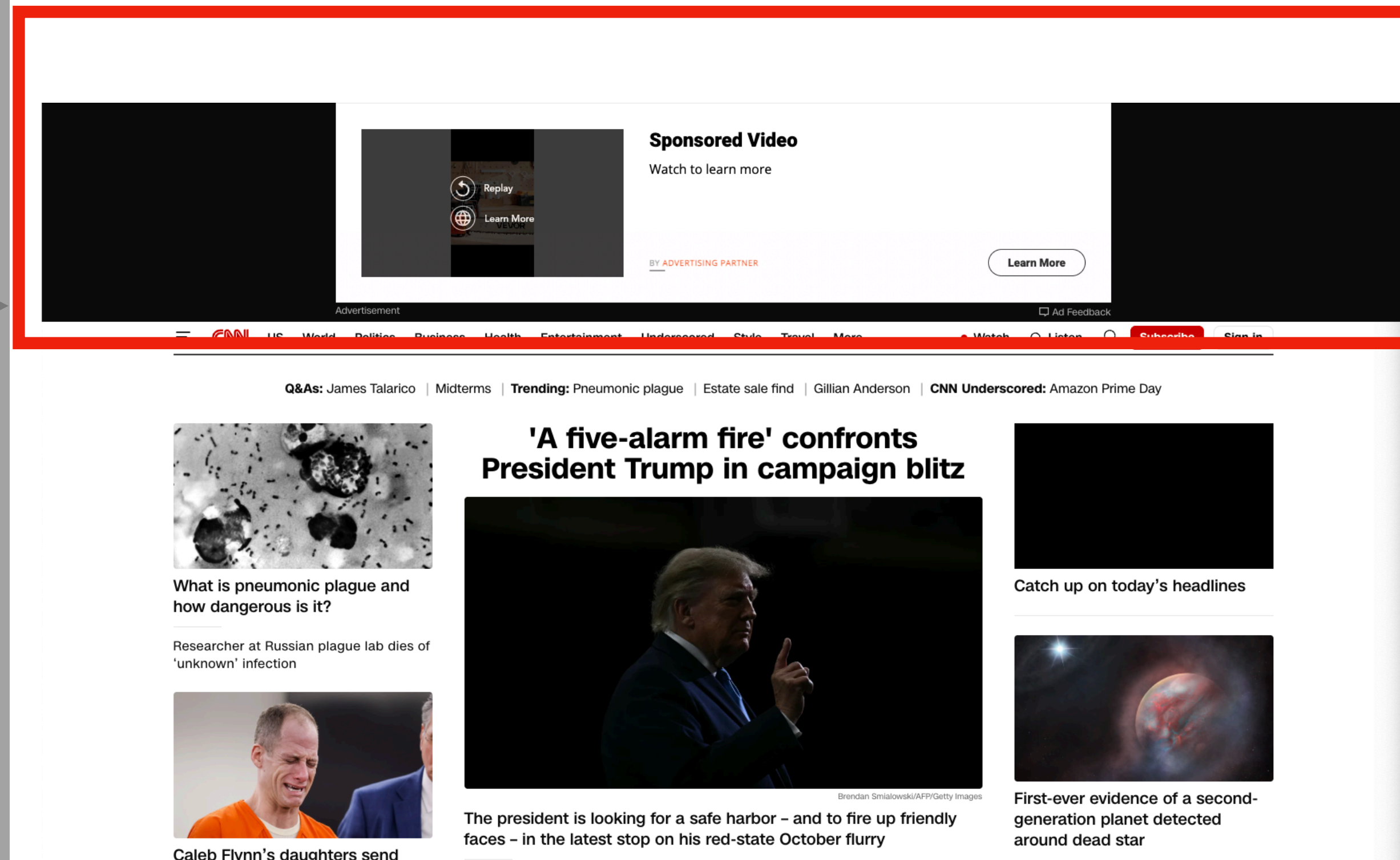
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cnn.com

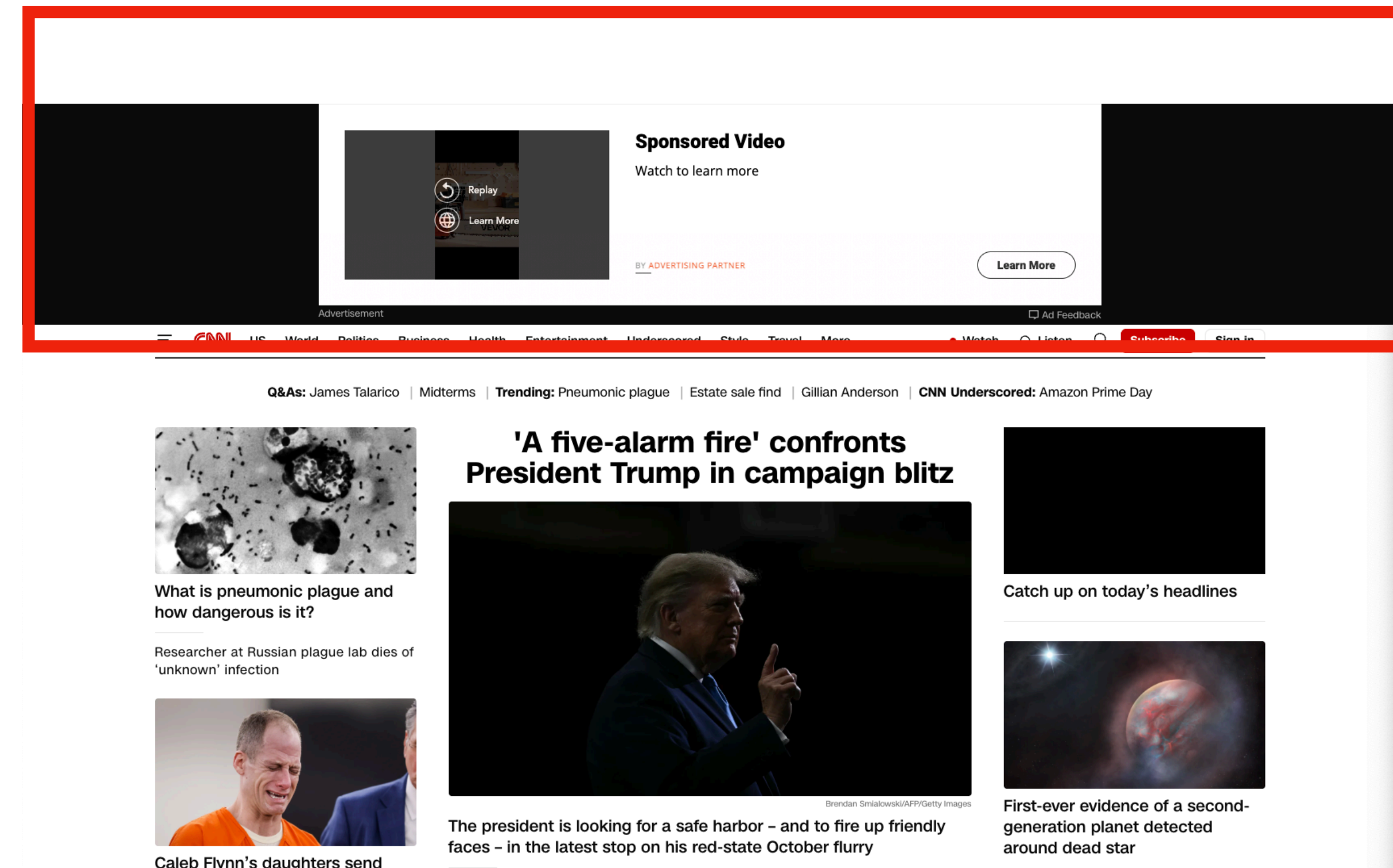
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GET google.com/ad

Cookie:
client_id=7582



cnn.com

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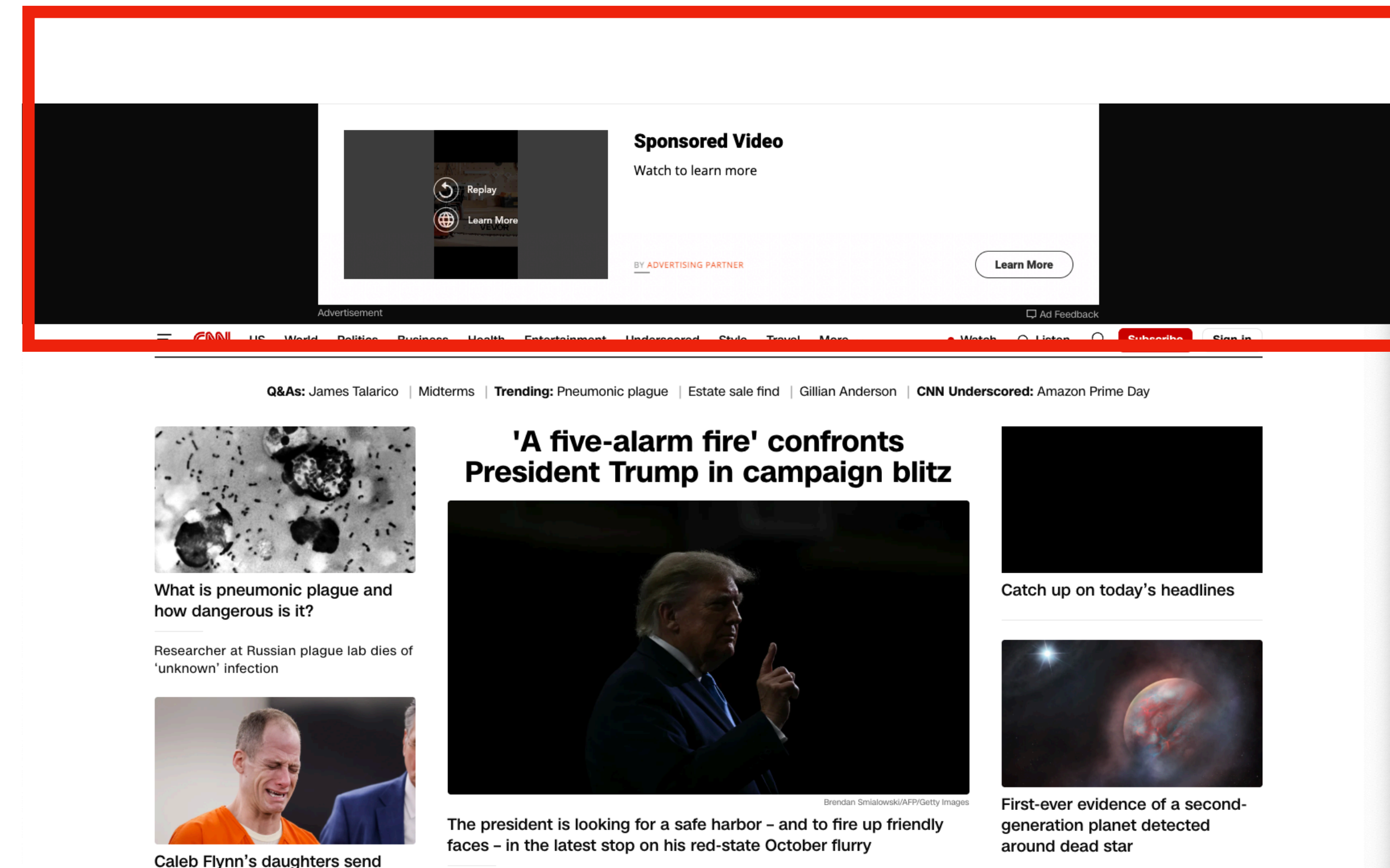


Client

GET google.com/ad

Cookie:
client_id=7582

Referer:
cnn.com



cnn.com

What does Google now know about me?

What does Google now know about me?

- They know I went to Google (and all my searches)
- They know I went to cnn.com
 - If I *click* on anything on cnn.com, they can know that too (assuming there's an ad on cnn.com)
- They know *when* I go anywhere on the Internet, assuming they have *any* kind of digital presence (e.g., an ad *requested* on the page)

Surely, Google can't be in that many places, right?

Surely, Google can't be in that many places, right?

Wrong

- Major companies have large presences on the web, and as a result, can see the majority of websites that you visit
- Google appears on **82.2%** of the Top 1M, primarily because of analytics and advertising services
- Facebook appears on **34.1%**, to enable social sharing + tracking

Company	Prevalence on Top 1M
Google	82.2%
Facebook	34.1%
Amazon	32.6%
Cloudflare	30.7%
Akamai	20.3%
MaxCDN	19.0%
Edgecast	17.9%
Fastly	15.5%
SoftLayer	11.8%
Twitter	11.2%

Third Party Tracking is Everywhere

- Third-party cookies — cookies sent during a browsing instance that are *not to the page you're visiting*
- Not just ads! How else can this type of tracking be enabled?
 - Images, invisible pixels, redirects... basically **anything that makes the browser make an HTTP request**

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- Not just ads! How else can this type of tracking be enabled?
 - Images, invisible pixels, redirects... basically **anything that makes the browser make an HTTP request**
- Third-party tracking has been, for most of the history Internet, the de-facto way to track and follow you around as you browse
 - It happens transparently (without your knowledge) and is **built into the web ecosystem by default**

Maybe we're not doomed?

- With a group, discuss the following question:
 - In what ways is this type of stateful, third-party tracking *limited* in its ability to track you?

Limitations of vanilla third-party tracking

- Each tracker only sees its own slice of the web
 - E.g., Facebook's tracking is "less comprehensive" than Google's tracking, just based on their vantage point
- Users might install extensions, or other software that block third-party requests altogether (some browsers do this, for example!)
- Users might periodically delete their cookies, forcing new `Set-Cookie` headers and therefore breaking linkages

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All of these have been cleverly bypassed by the ecosystem over time.

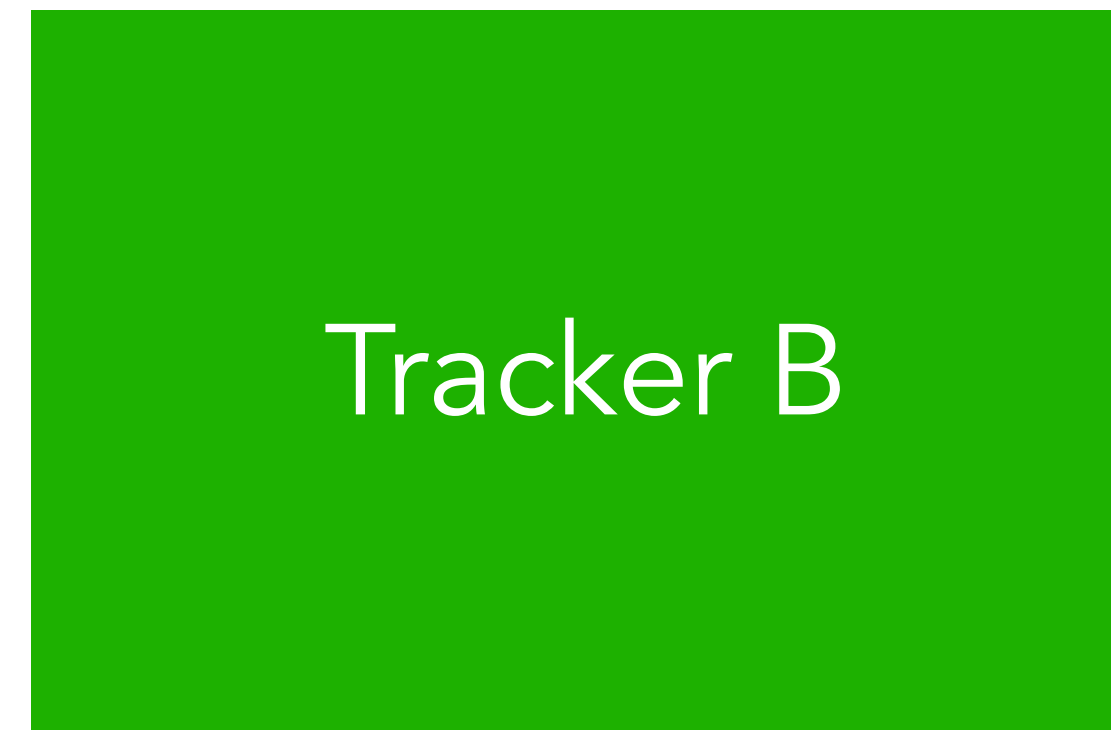
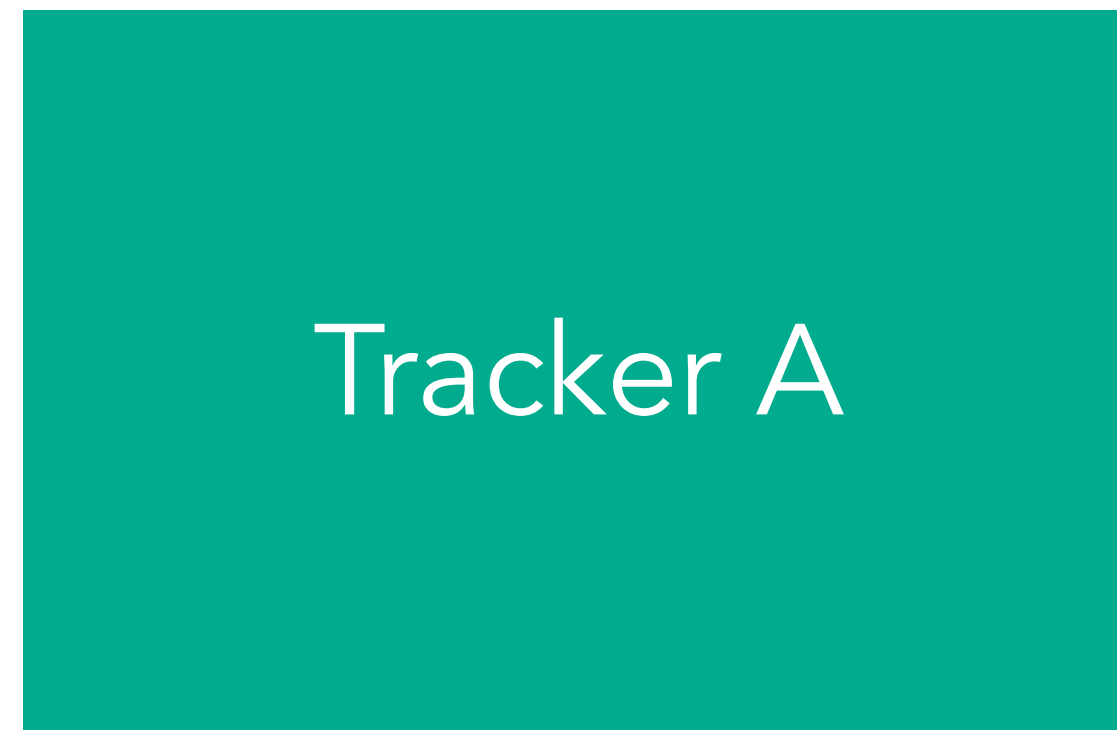
Bypassing Cookie Protections

A tale of two trackers

- Problem — TrackerA and TrackerB only see their own slice of the world, but they'd like to see more

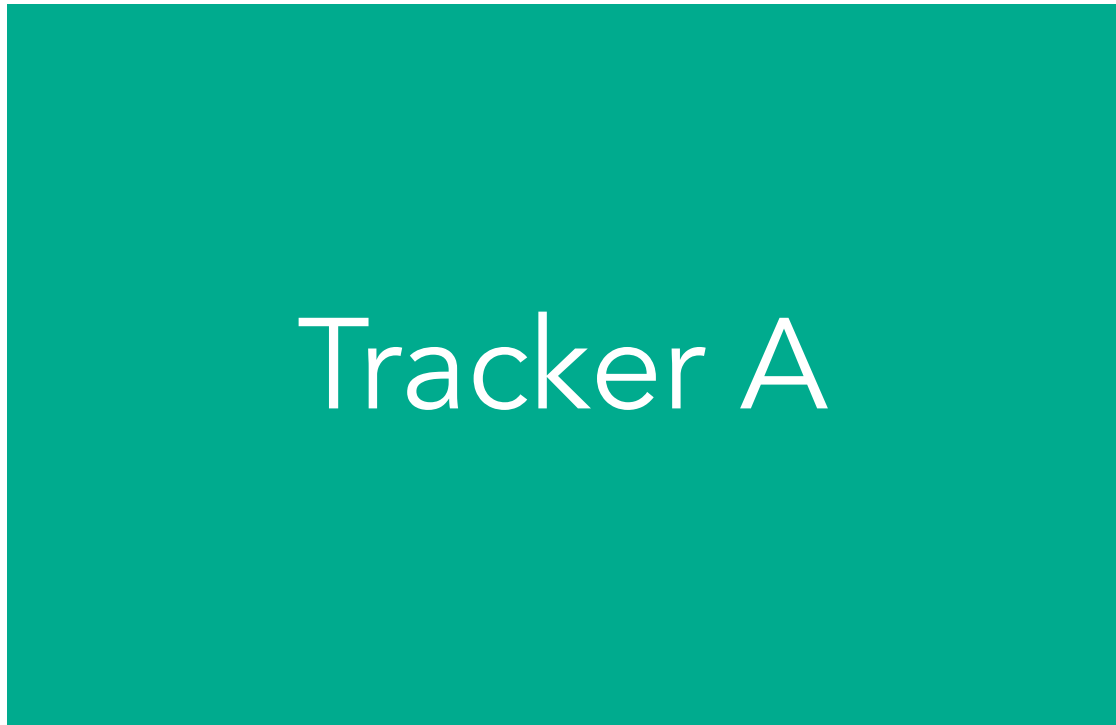
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Tracker A

- NYTimes
- CNN
- HuffingtonPost

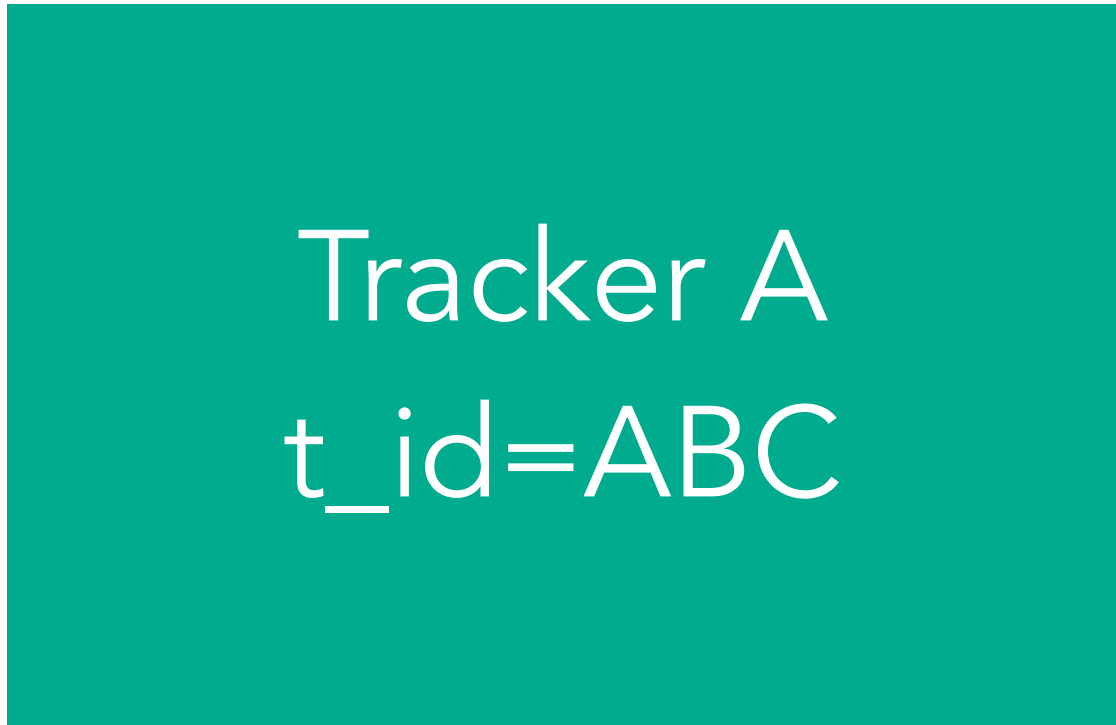


Tracker B

- HackerNews
- TechCrunch
- Wired

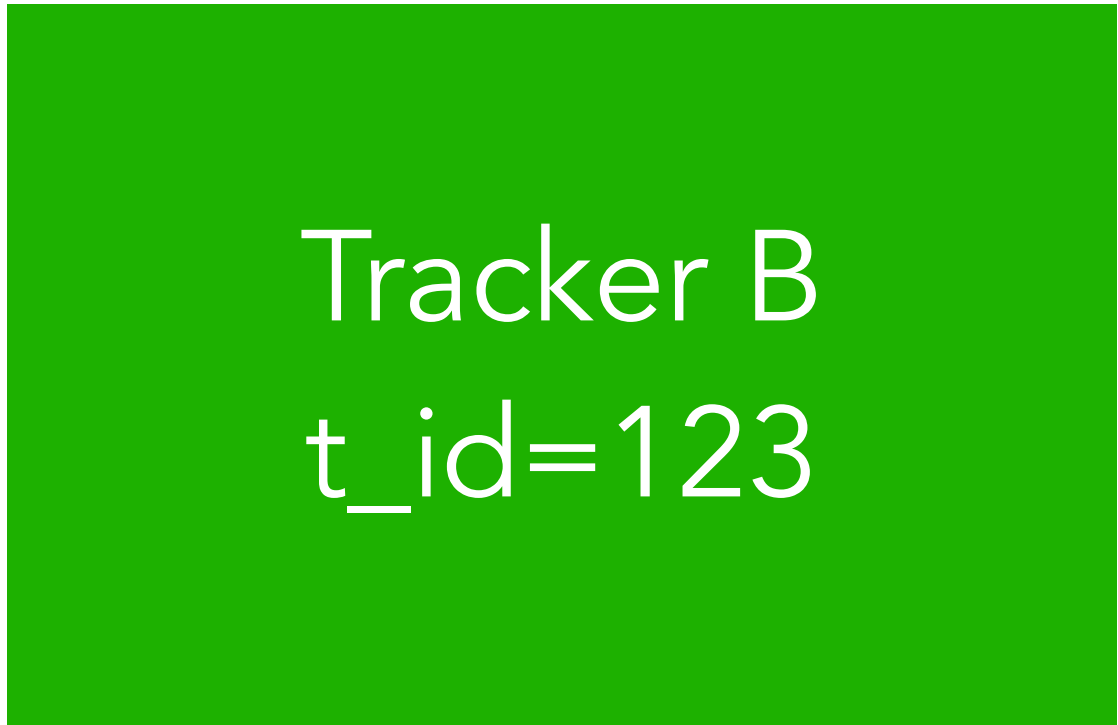
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Tracker A
t_id=ABC

- NYTimes
- CNN
- HuffingtonPost



Tracker B
t_id=123

- HackerNews
- TechCrunch
- Wired

TrackerA

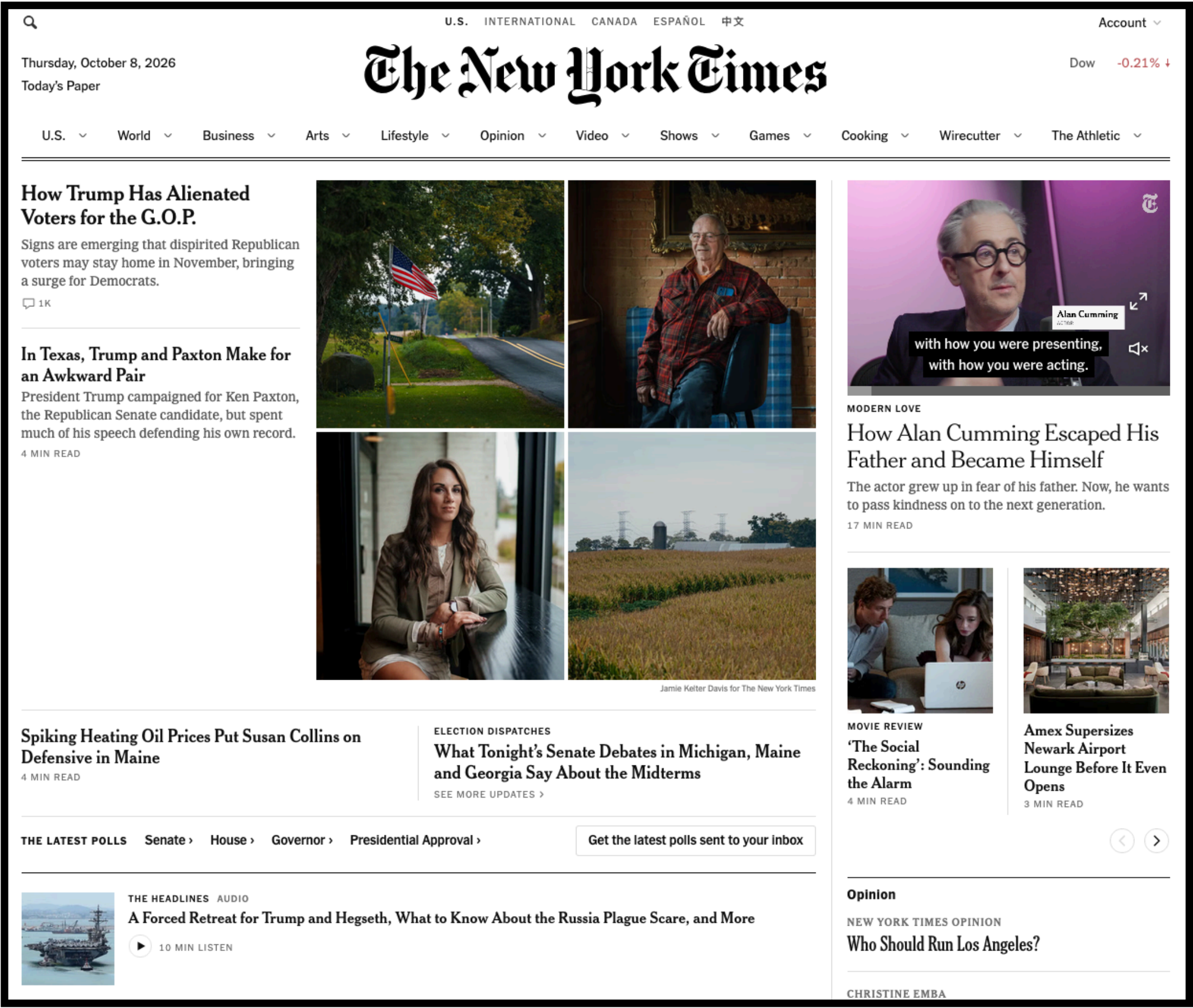


TrackerA

GET nytimes.com/

NYTimes Response

Client



TrackerA

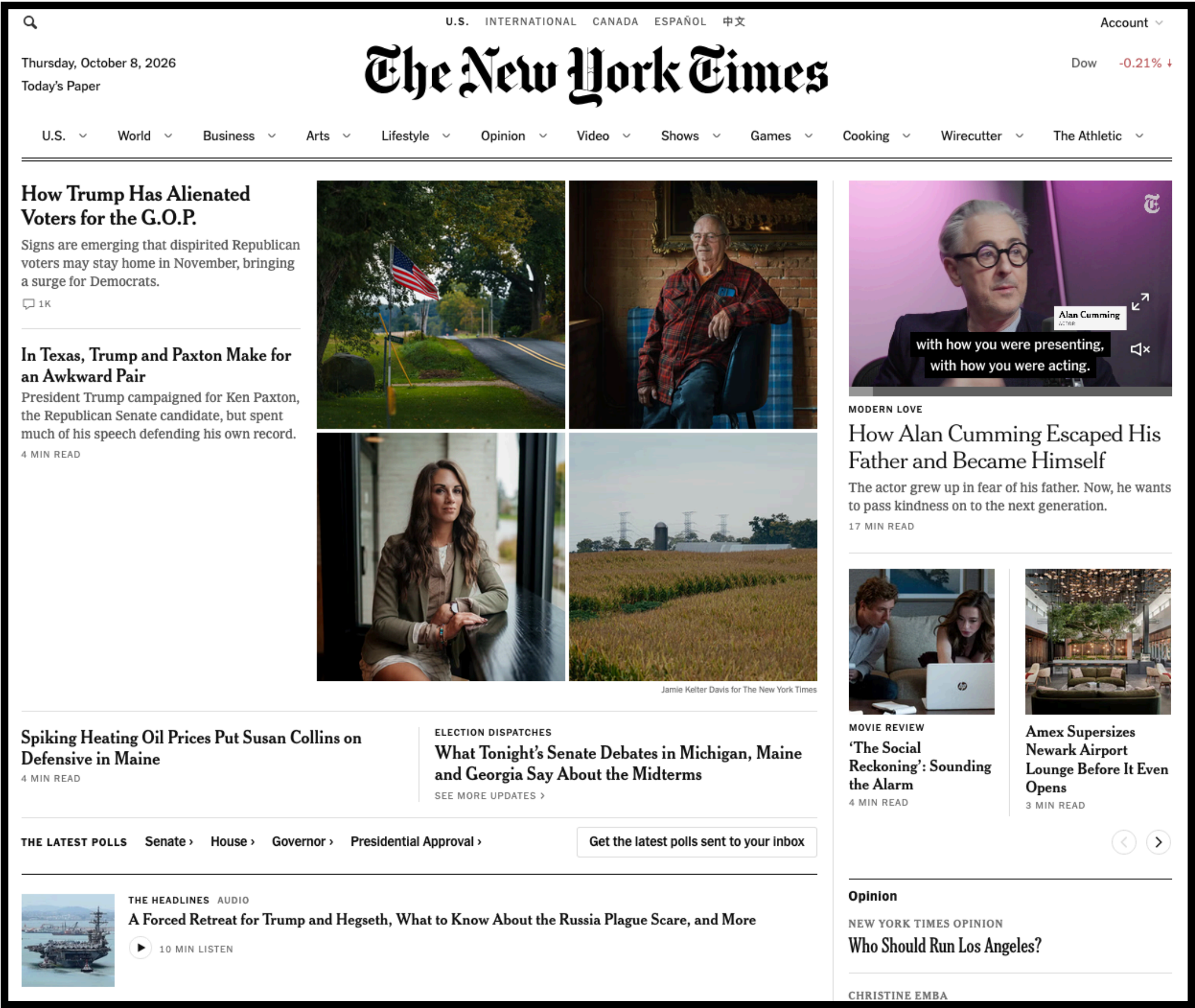
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NYTimes Response

GET TrackerA



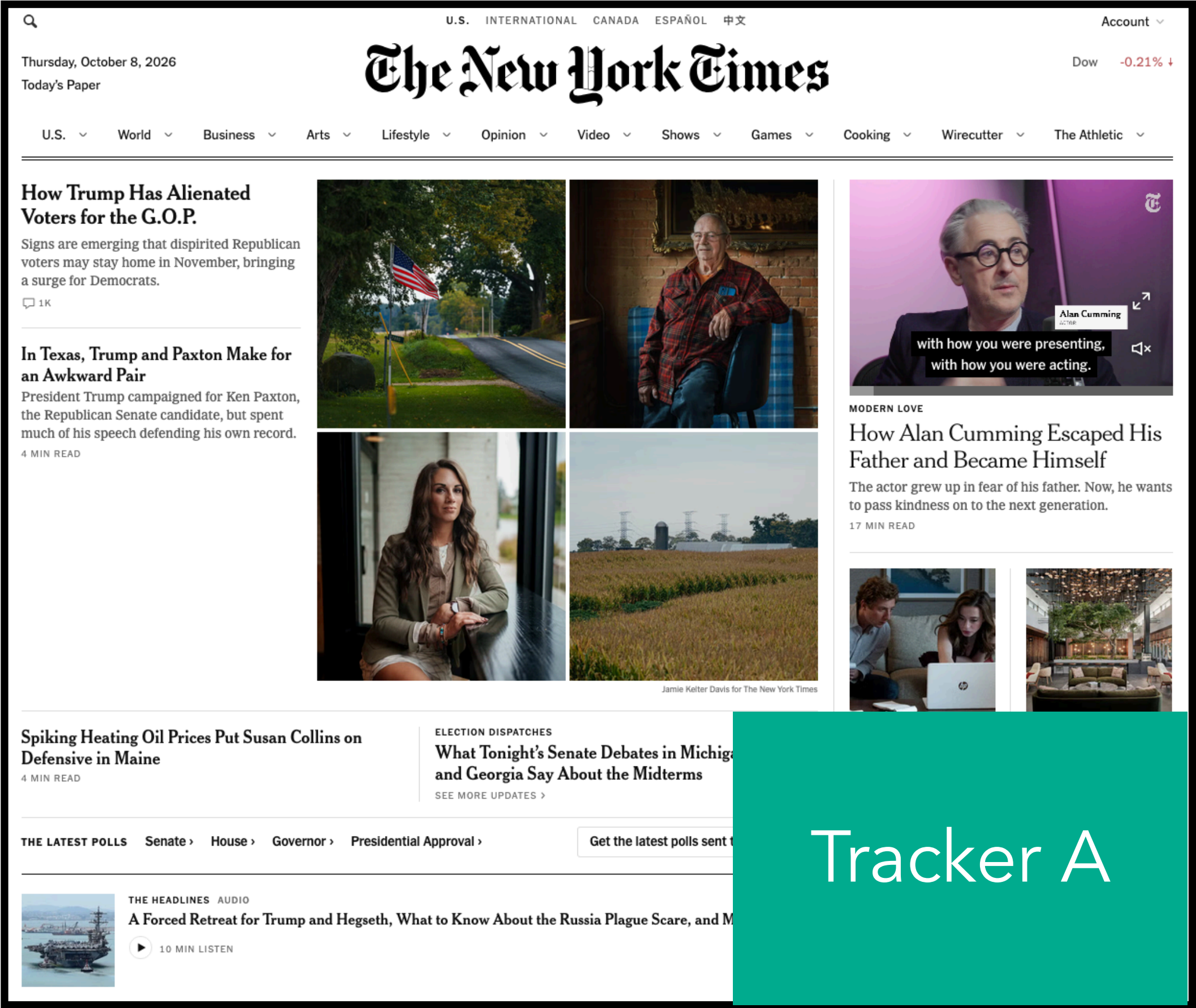
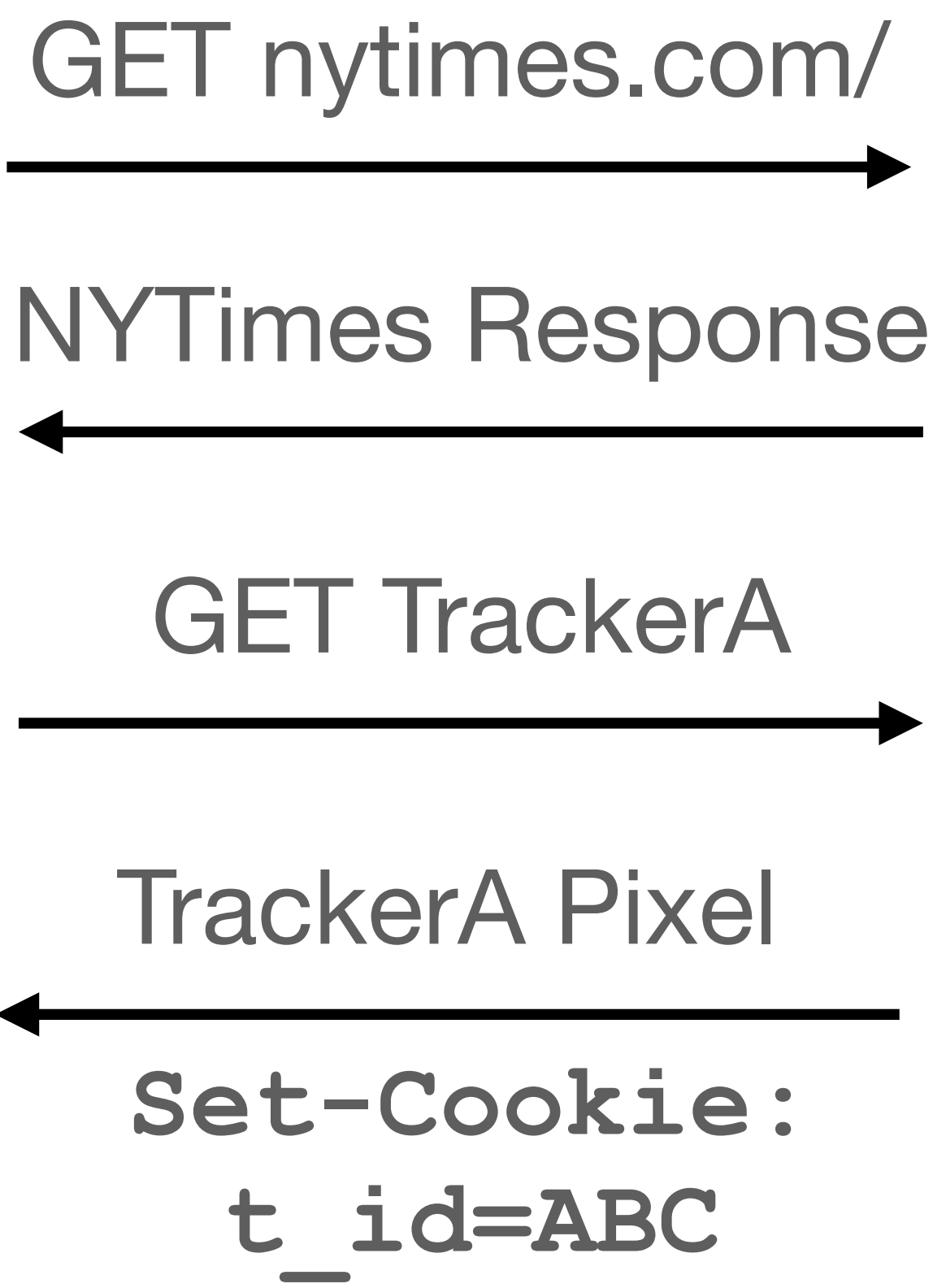
Client



TrackerA



Client



TrackerB

GET HackerNews

HN Response

GET TrackerB

TrackerB Pixel

Set-Cookie:
t_id=123

Y

Hacker News new | past | comments | ask | show | jobs | submit

1. Beauty in DVD Menus (vale.ocks)
74 points by speckx 1 hour ago | hide | 53 comments

2. Show HN: I've been paying for a rural Tanzanian's education for 10 years (tanzaniaeducationproject.org)
44 points by lukehandcool 31 minutes ago | hide | 13 comments

3. "Math 2.0" will need to value mathematical progress more holistically (mathstodon.xyz/tao)
475 points by ent101 9 hours ago | hide | 461 comments

4. The Slow Formation of Durable Software (dancohen.org)
124 points by benbreen 4 hours ago | hide | 39 comments

5. Telnet BBS Guide (telnetbbsguide.com)
51 points by kmstout 2 hours ago | hide | 22 comments

6. Claude Haiku 5.5 (anthropic.com)
973 points by sfgtbor 21 hours ago | hide | 453 comments

7. OpenAI Withdraws 3 Math Papers (github.com/openai)
278 points by theemathas 7 hours ago | hide | 231 comments

8. The 15-year search for a band that charted once and vanished (shahidhussain.com)
260 points by shahidhussain 11 hours ago | hide | 96 comments

9. I gave Opus 5.5 one prompt and six hours to visualize Invisible Cities (quesma.com)
169 points by stared 3 hours ago | hide | 100 comments

10. Time Travel in Braid (2015) (qntm.org)
41 points by Arianule 4 hours ago | hide | 13 comments

11. Living off-grid: Hundred Rabbits (100r.ca)
309 points by Muhammad523 13 hours ago | hide | 119 comments

12. Margaret Hamilton has died (news.mit.edu)
1913 points by muglug 17 hours ago | hide | 212 comments

13. GPT-6 and Intelligent UI for everyone (openai.com)
705 points by joshuawright11 21 hours ago | hide | 414 comments

14. How did Rosalind Franklin miss the helix in her iconic DNA image? She didn't (science.org)
250 points by pavel_ishin 16 hours ago | hide | 83 comments

15. 'Jonathan' is the oldest land animal on Earth (404media.co)
214 points by gumby 16 hours ago | hide | 121 comments

16. Cleo (Mathematician) (wikipedia.org)
214 points by djoldman 15 hours ago | hide | 41 comments

17. The Mathocalypse (scottaaronson.blog)
320 points by 6bitquant 19 hours ago | hide | 331 comments

18. Anne Carson wins Nobel Prize in literature 2026 (theguardian.com)
85 points by sonabinu 3 hours ago | hide | 6 comments

19. Float and integer arithmetic follow two different paradigms (pkh.me)
16 points by ibobev 3 hours ago | hide | 4 comments

20. 'The eels were climbing out of the water to die' (theguardian.com)
46 points by speckx 1 hour ago | hide | 16 comments

21. Show HN: Bigwords.page - Turn any screen into a sign. The URL is the app (bigwords.page)
602 points by SpeakingOfBrad 23 hours ago | hide | 145 comments

Tracker B

Client

72

TrackerA 🤝 TrackerB

- TrackerA and TrackerB have different identifiers for the same user!
- So how can TrackerA and TrackerB work together to use their vantage points to build a complete picture of each user?
- They can independently agree to **synchronize** cookies automatically when the browser makes requests — through a tactic called *Cookie Syncing*

TrackerA 🤝 TrackerB



Client

GET TrackerA



Tracker A

Tracker B

TrackerA 🤝 TrackerB



Client

GET TrackerA

302 trackerb.com/sync?t_id=ABC

Tracker A

Tracker B

TrackerA 🤝 TrackerB



Client

GET TrackerA

302 trackerb.com/sync?t_id=ABC

GET trackerb.com/sync?t_id=ABC
Referer: TrackerA

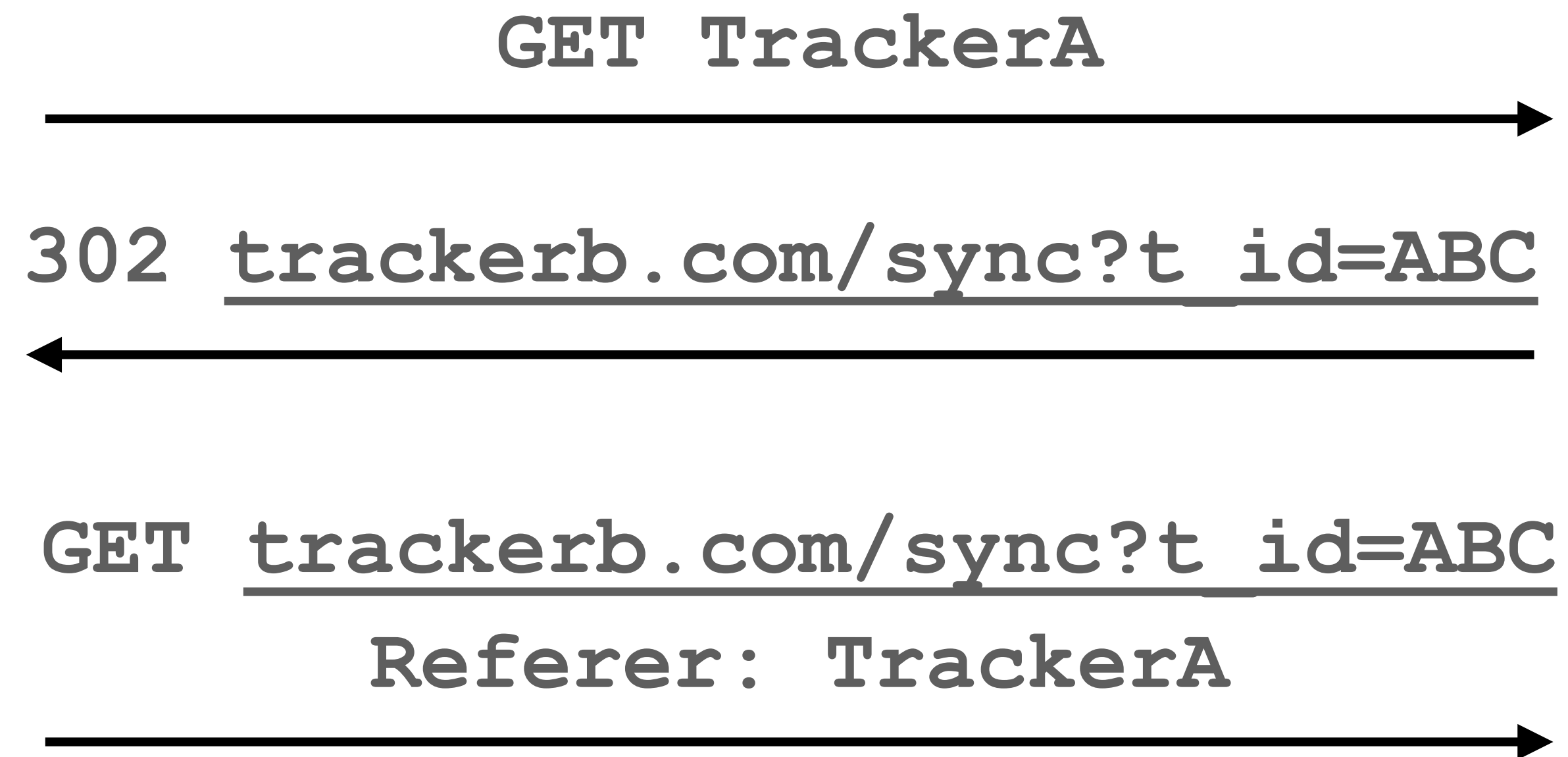
Tracker A

Tracker B

TrackerA 🤝 TrackerB



Client



**TrackerB now knows TrackerA's
cookie identifier, and can match
in a DB on their backend!**

Tracker A

Tracker B

What just happened?

- TrackerA and TrackerB had disparate **cookies** for the same browser
- But after cookie syncing, **both** now have a linkage between what “TrackerA” calls this person and what “TrackerB” calls this person
- Question: Is cookie syncing restricted to just two trackers?

What just happened?

- TrackerA and TrackerB had disparate **cookies** for the same browser
- But after cookie syncing, **both** now have a linkage between what “TrackerA” calls this person and what “TrackerB” calls this person
- Question: Is cookie syncing restricted to just two trackers? **No!**
- Why does this matter?
 - It defeats browser protections put in place by cookie scoping
 - Your data is traveling further than you might expect
 - It relies on underlying browser feature that simply *has* to work for the web to function (redirects)

Cookie Syncing in the Wild

- [See Demo](#)

Defeating Cookie Syncing – Partitions

- Many browsers (Safari, Firefox, Brave) have gotten fed up with all this tracking, and implemented **cookie partitioning**
- For example — Firefox calls this Total Cookie Protection (other browsers call it other things), released in **2021**



Introducing Total Cookie Protection in Standard Mode

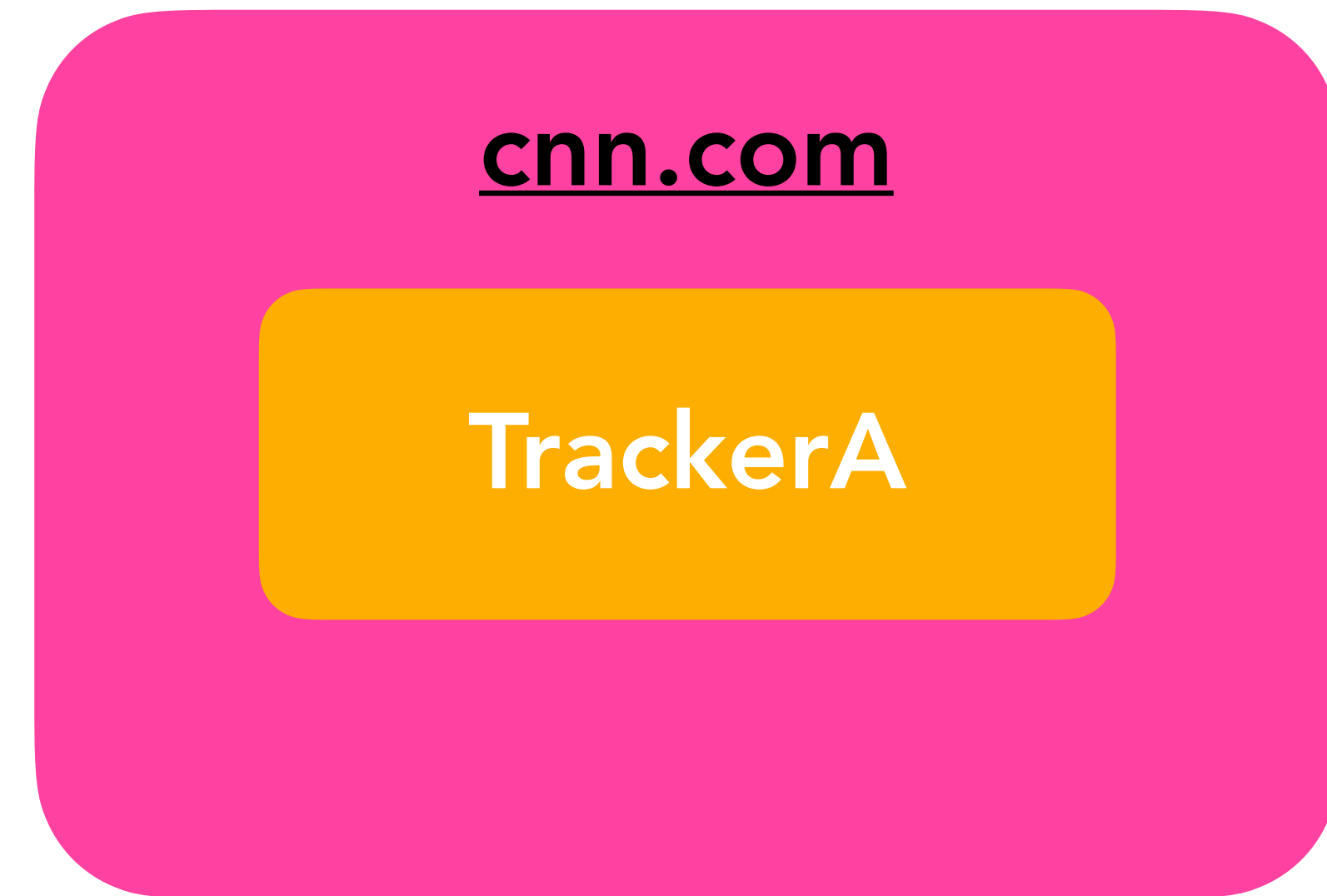
Firefox  Last updated: 5/20/25  73% of users voted this helpful

Firefox users now have access to Total Cookie Protection by default. Total Cookie Protection is an advancement built into [Enhanced Tracking Protection](#) that works by maintaining a separate cookie jar for each website you visit.

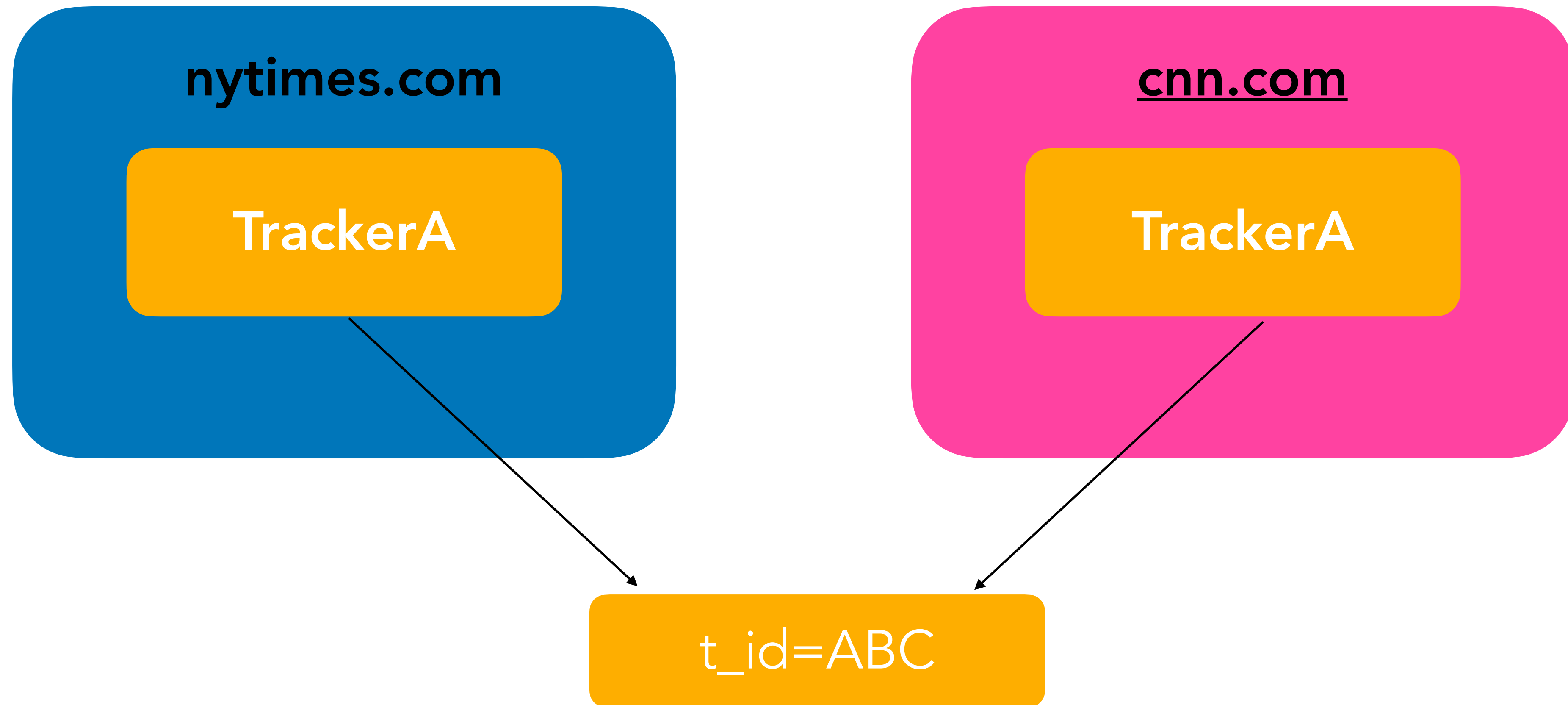
Cookie Partitions

- Instead of having cookie scope *solely* determined by “which domain sets the cookie” — have it also scoped by **the top-level page**
- In order for a browser to send a cookie, it must match **both** the top-level page *and* the third-party domain

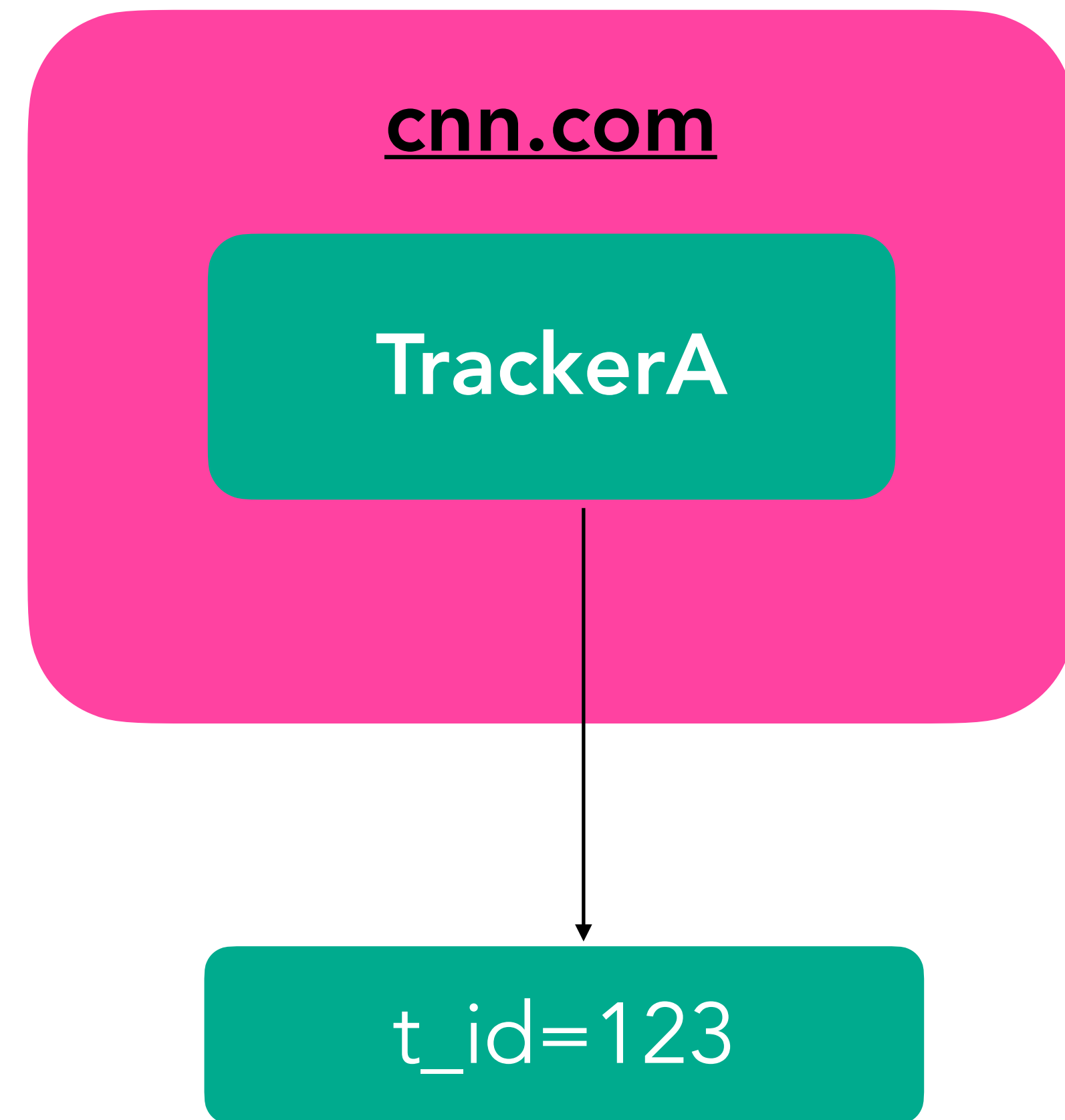
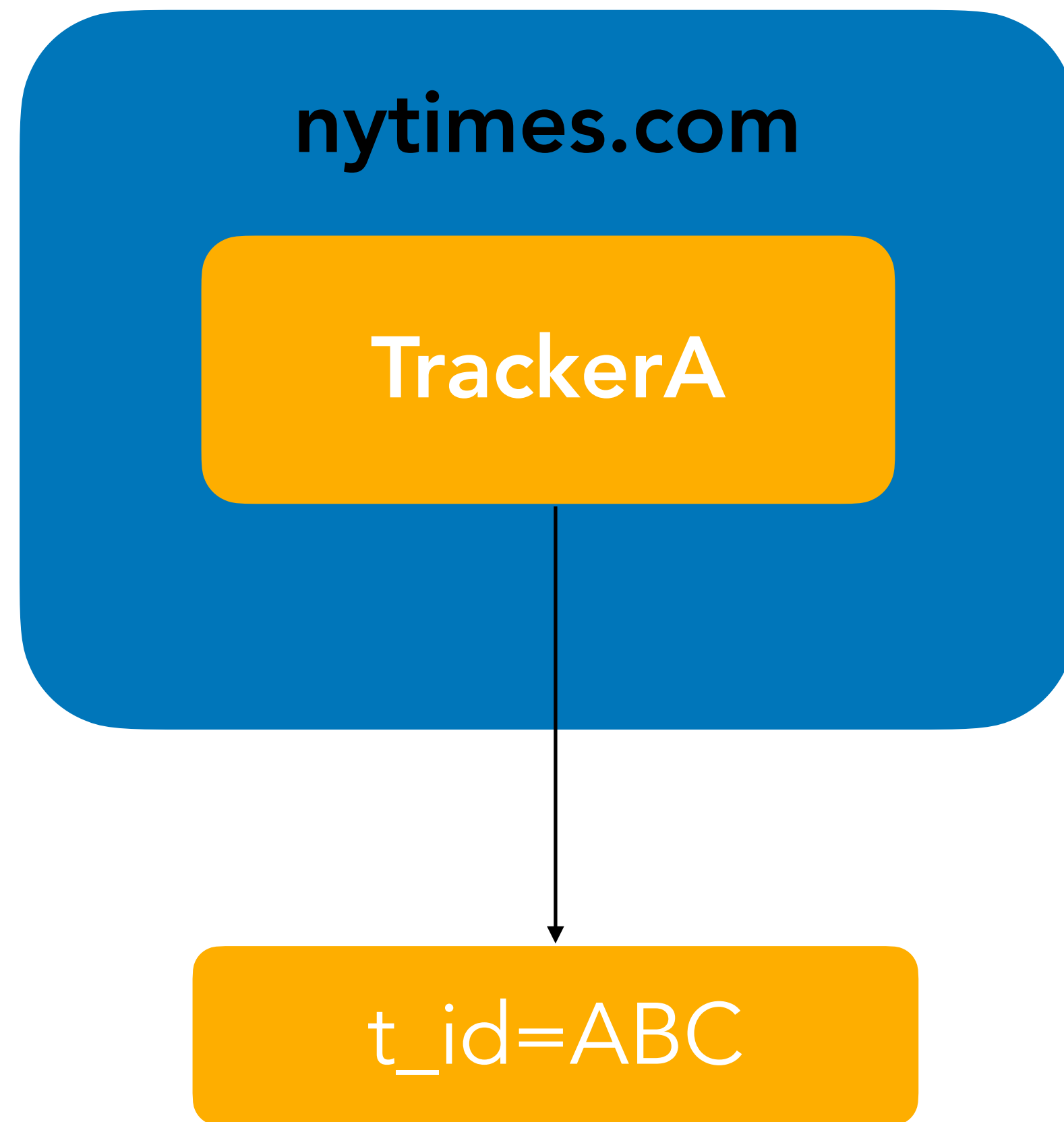
Third-Party Tracking



Third-Party Tracking



Partitioned Cookies

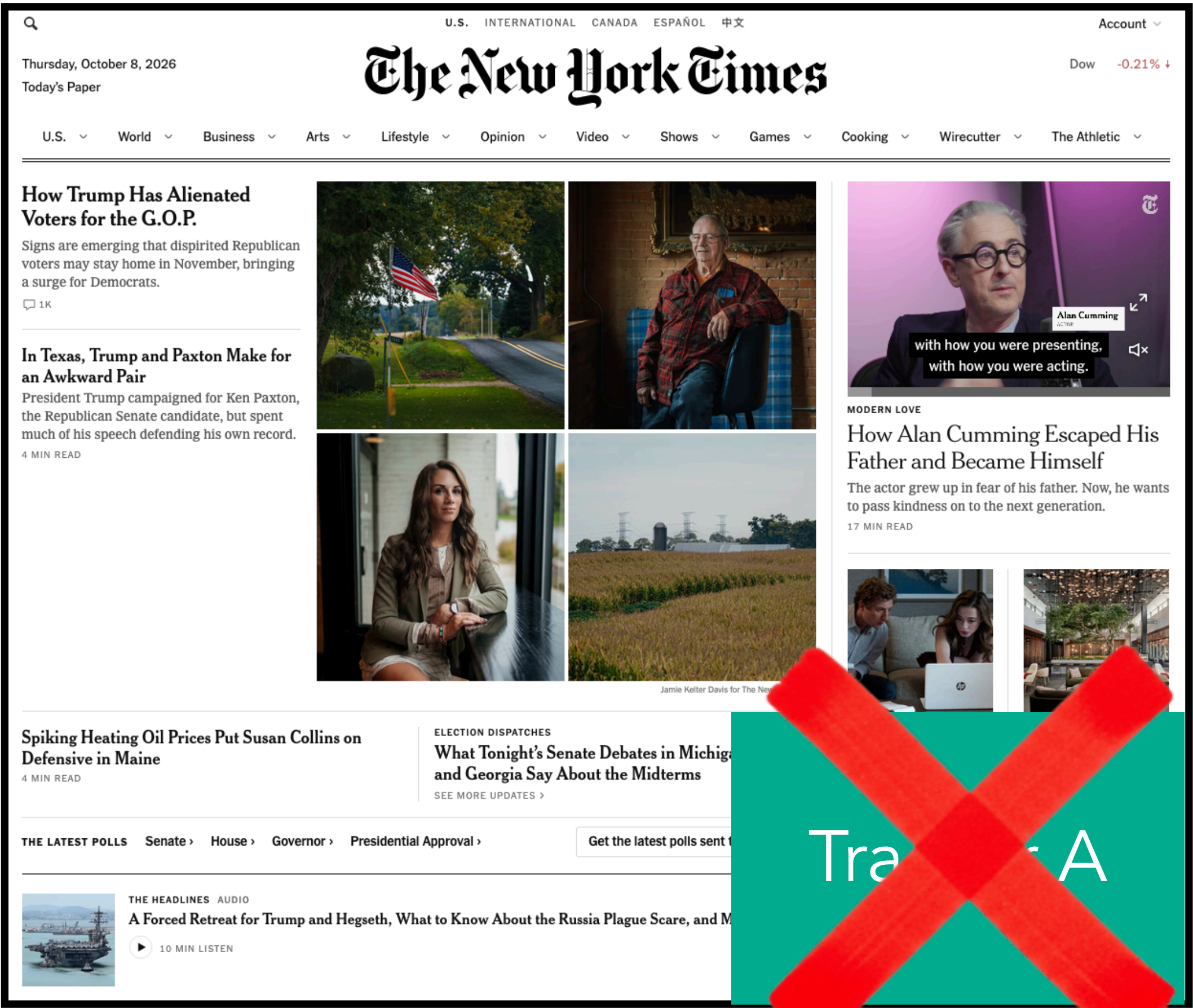
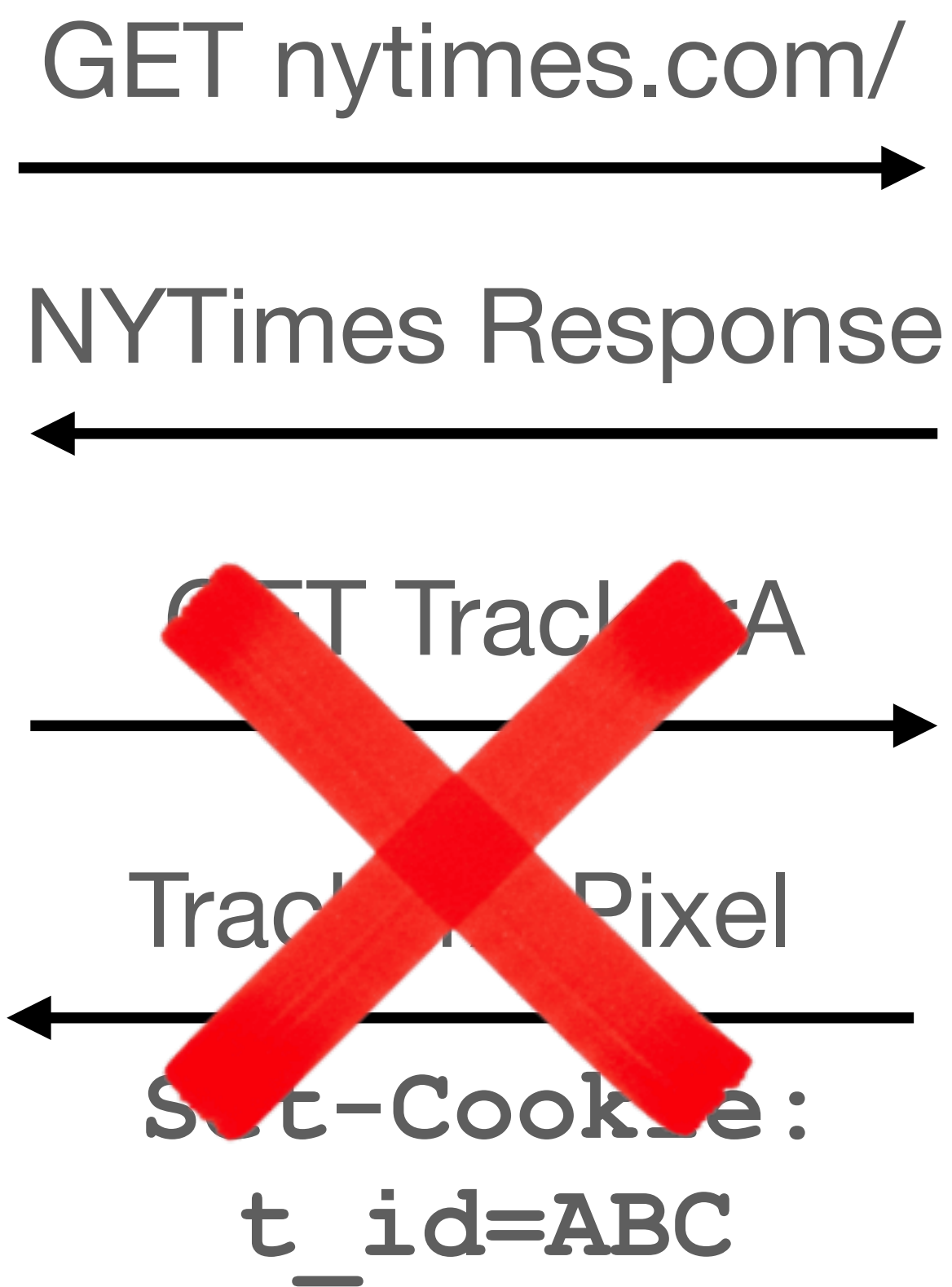


Why do Cookie Partitions Defeat Cookie Syncing?

- A tracker's *own* identifier is unique across different page visits, making cookie syncing *between* trackers essentially useless
- Cookie Partitioning is a **default** in most browsers!

Another way of beating cookie syncing... block third-parties

- The browser can automatically block third-party cookies from being sent!



Most browsers these days disable third-party cookies



Firefox



Brave



Safari

Most browsers these days disable third-party cookies



Firefox



Brave



Safari

Who is conspicuously missing from this list?

TECHNOLOGY

After 4 years, Google backtracks on its discontinuation of third-party cookies

JULY 27, 2024 · 7:51 AM ET

HEARD ON [WEEKEND EDITION SATURDAY](#)



Google reneges on plan to remove third-party cookies in Chrome

By [Aimee Picchi](#)

Updated on: July 22, 2024 / 5:45 PM EDT / CBS News

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Google Breaks Promise to Block Third-Party Cookies

BY [LENA COHEN](#) | AUGUST 2, 2024

Chrome is so sad they can't block third-party cookies



"At the same time, we recognize this transition requires significant work by many participants and will have an impact on publishers, advertisers, and everyone involved in online advertising."

Google Chrome is Keeping the Third-Party Tracking Dream Alive



Google implements partitioned cookies (if you turn them on)

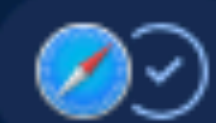
Partitioned

Cookies Having Independent Partitioned State (CHIPS)



Baseline 2025

NEWLY AVAILABLE



Since December 2025, this feature works across the latest devices and browser versions. This feature might not work in older devices or browsers.

Want more browser support for this feature? [Tell us why.](#)

[See full compatibility](#)

[Learn more](#)

Trackers still need to contend with third-party cookies going away

- Why?
 - If you use a browser where they can't track you
 - If you install an *extension* which stymies third-party blocking (e.g., most Ad blockers)
- So... what have trackers done about it?

Bounce Tracking

Sponsored Video

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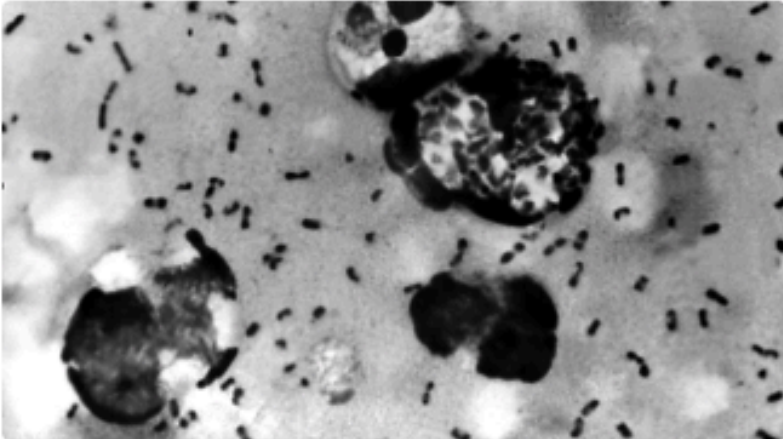
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Search

Subscribe

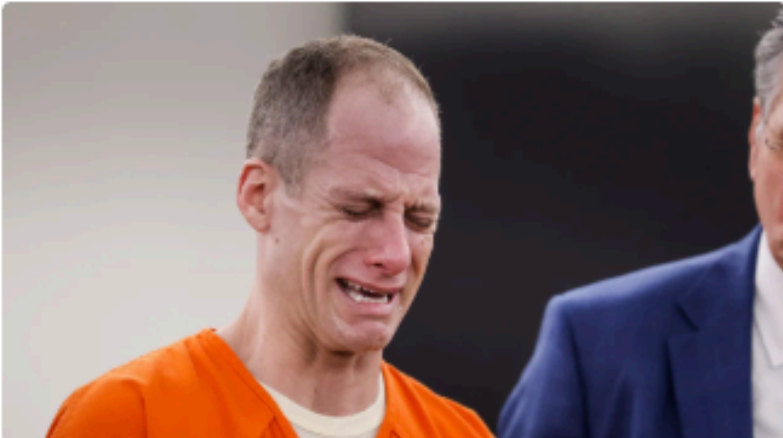
Sign in

Q&As: James Talarico | Midterms | Trending: Pneumonic plague | Estate sale find | Gillian Anderson | CNN Underscored: Amazon Prime Day



What is pneumonic plague and how dangerous is it?

Researcher at Russian plague lab dies of 'unknown' infection



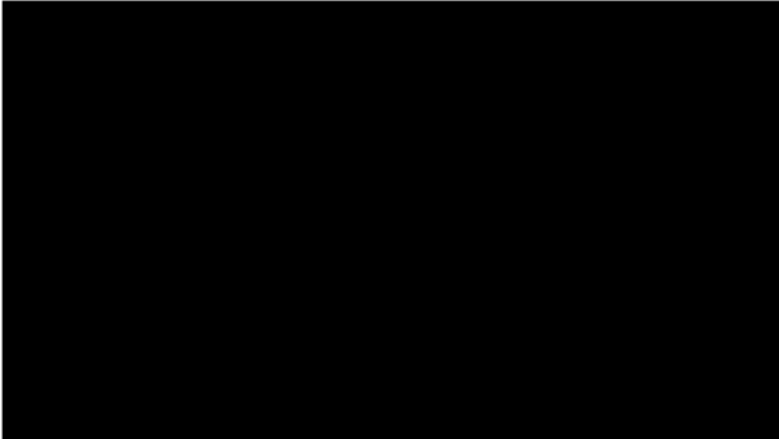
Caleb Flynn's daughters send

'A five-alarm fire' confronts President Trump in campaign blitz



Brendan Smialowski/AFP/Getty Images

The president is looking for a safe harbor – and to fire up friendly faces – in the latest stop on his red-state October flurry



Catch up on today's headlines



First-ever evidence of a second-generation planet detected around dead star

If I click on this link, where will it go?

Sponsored Video

Watch to learn more

BY ADVERTISING PARTNER

Learn More

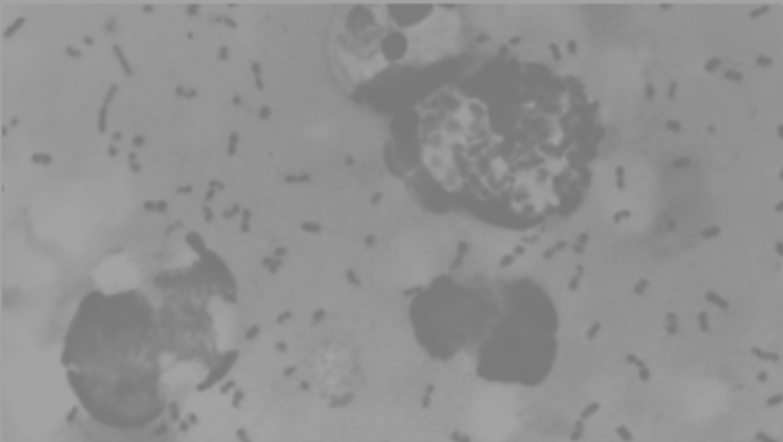
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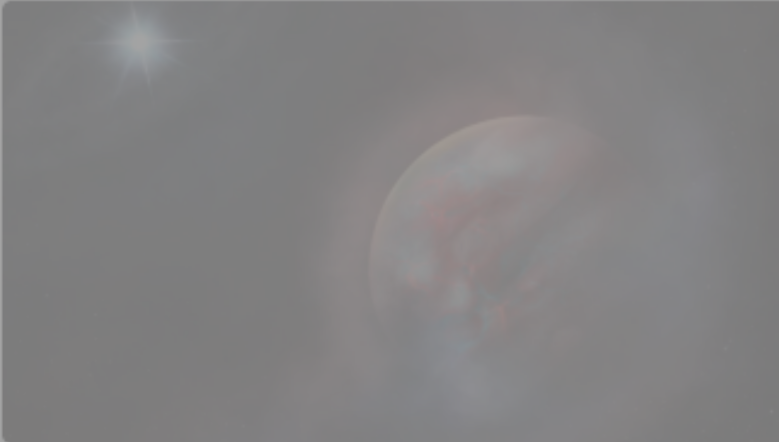
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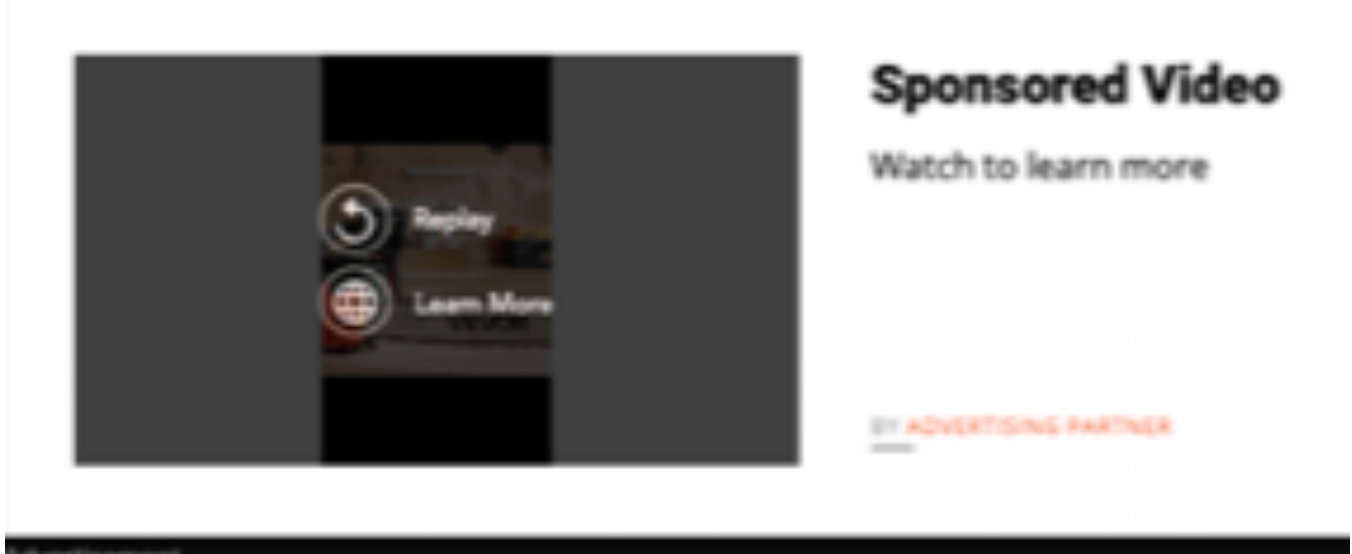
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Bounce Tracking



Client

GET ad.com/?
redirect=tracker.com



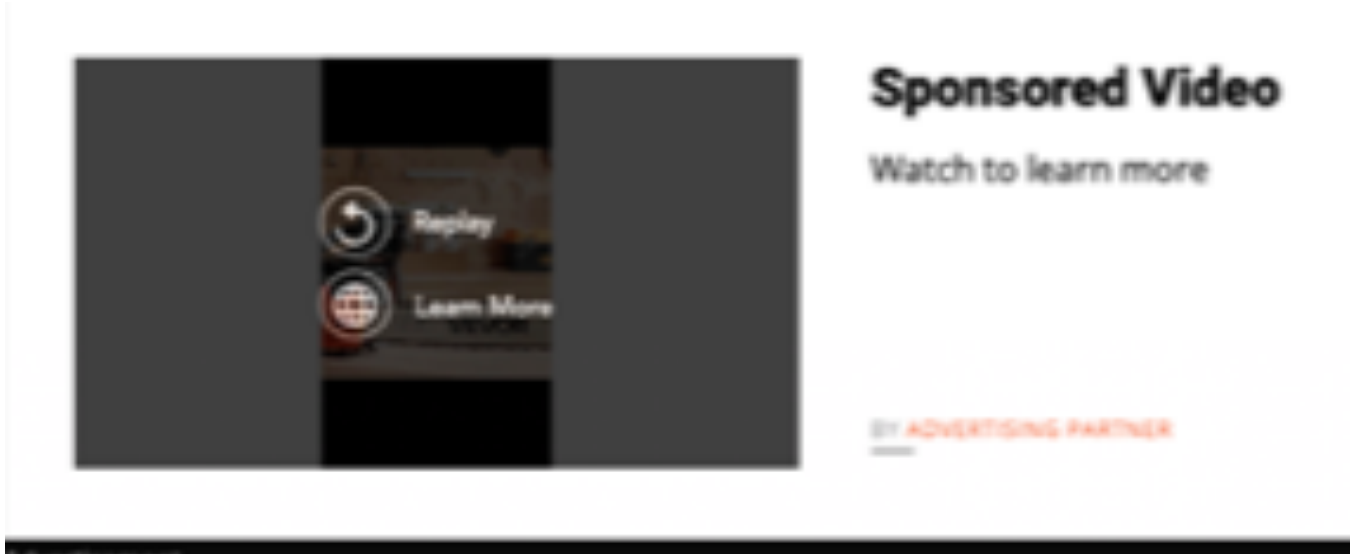
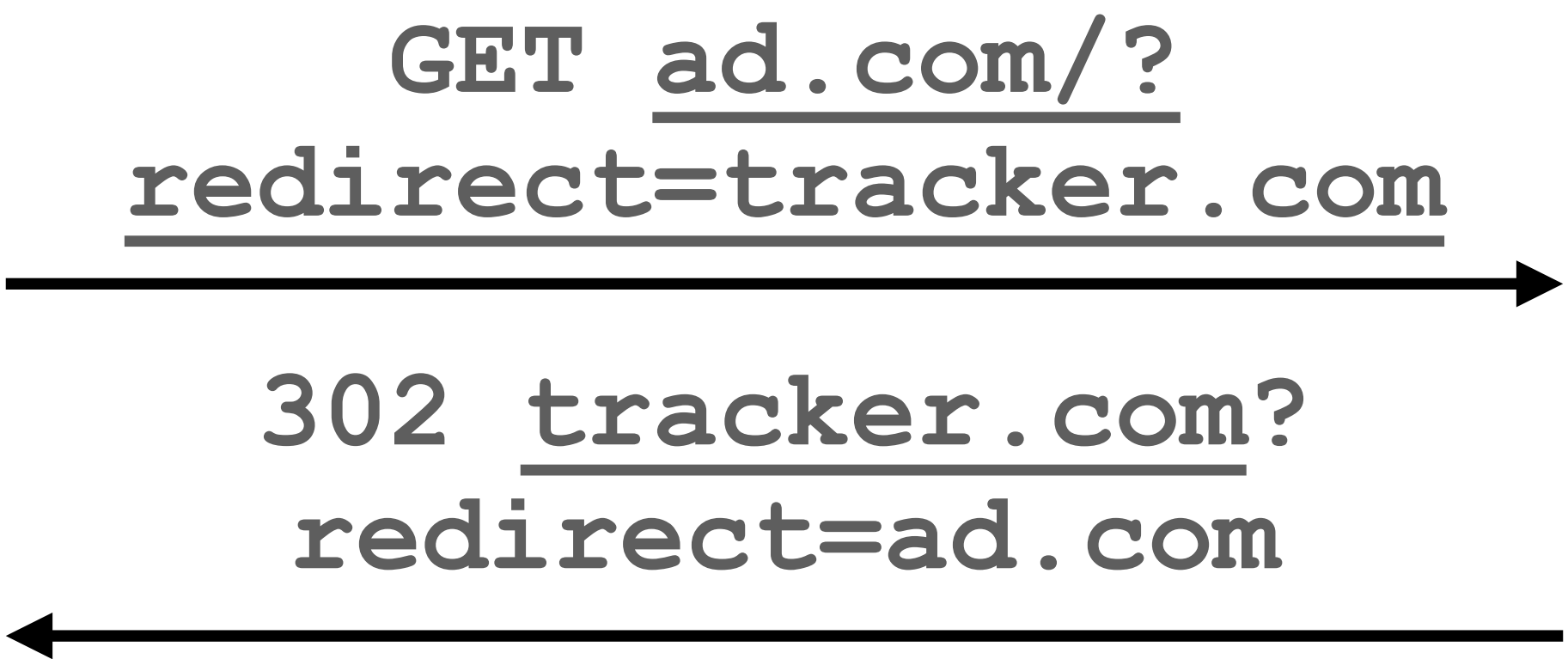
tracker.com

ad.com

Bounce Tracking



Client



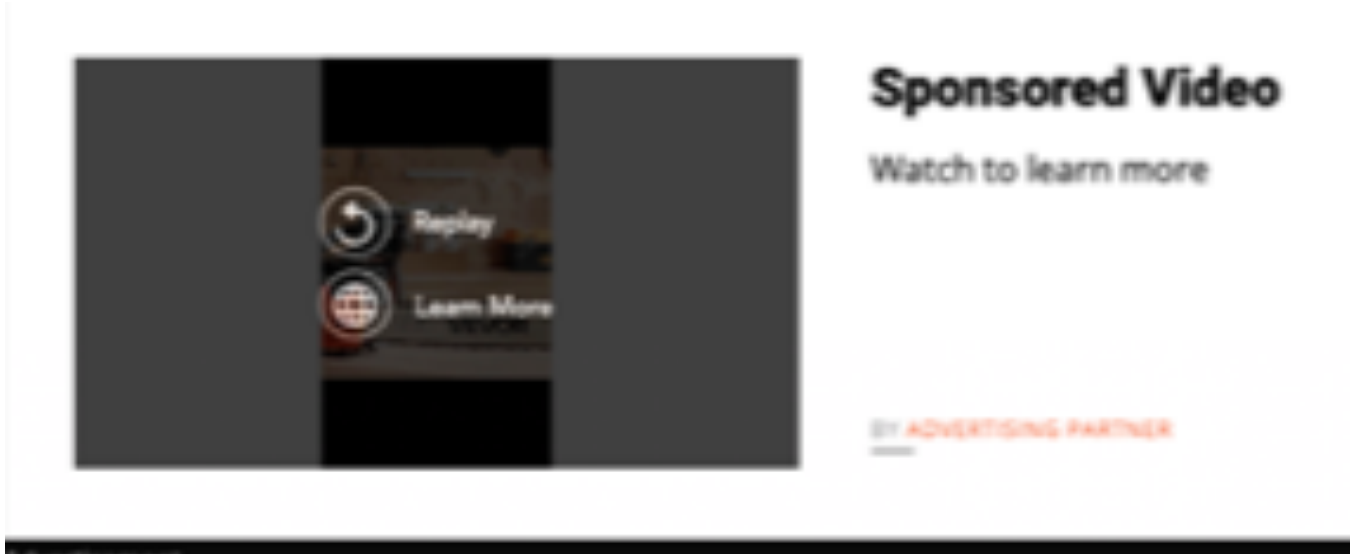
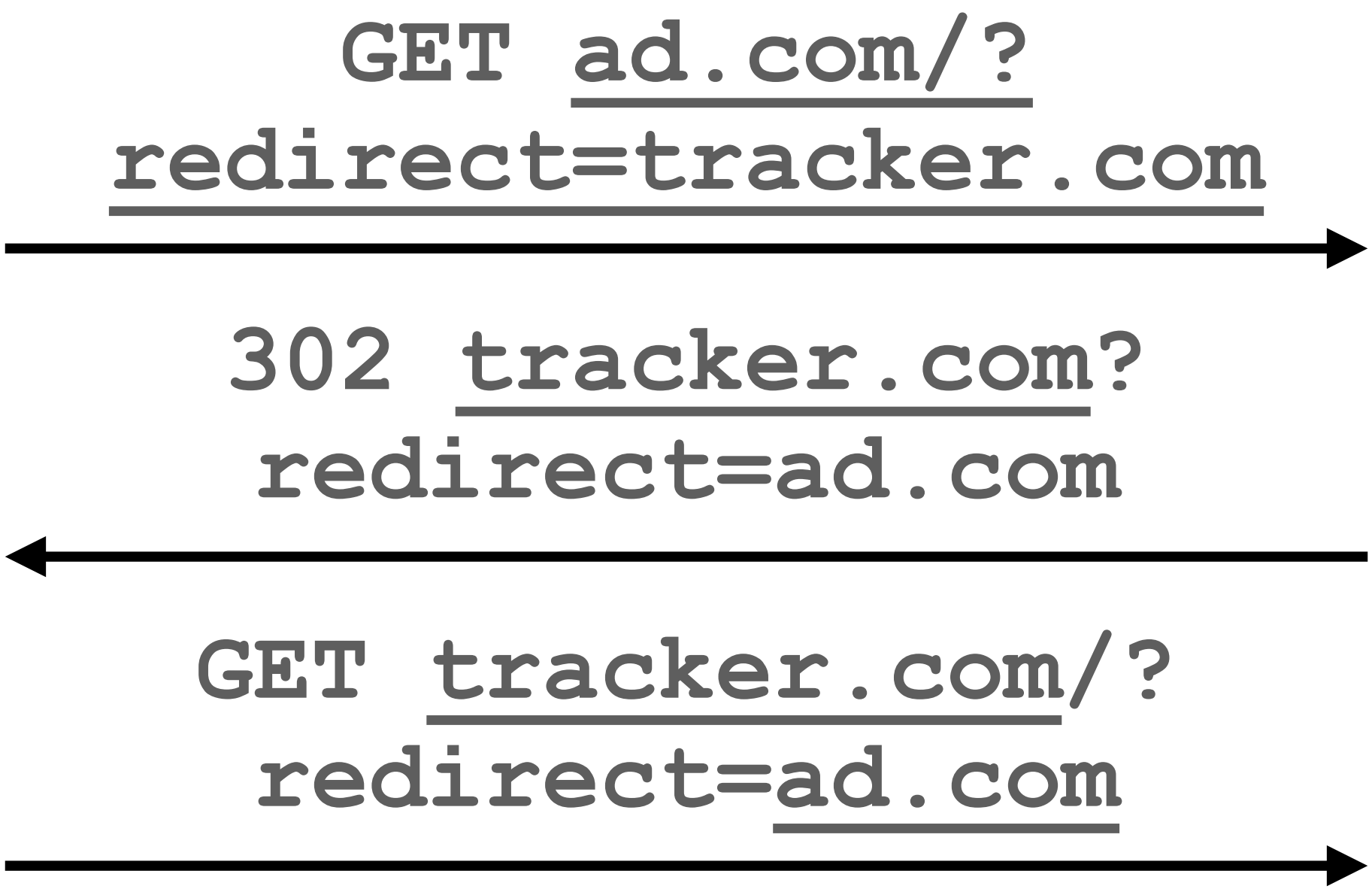
tracker.com

ad.com

Bounce Tracking



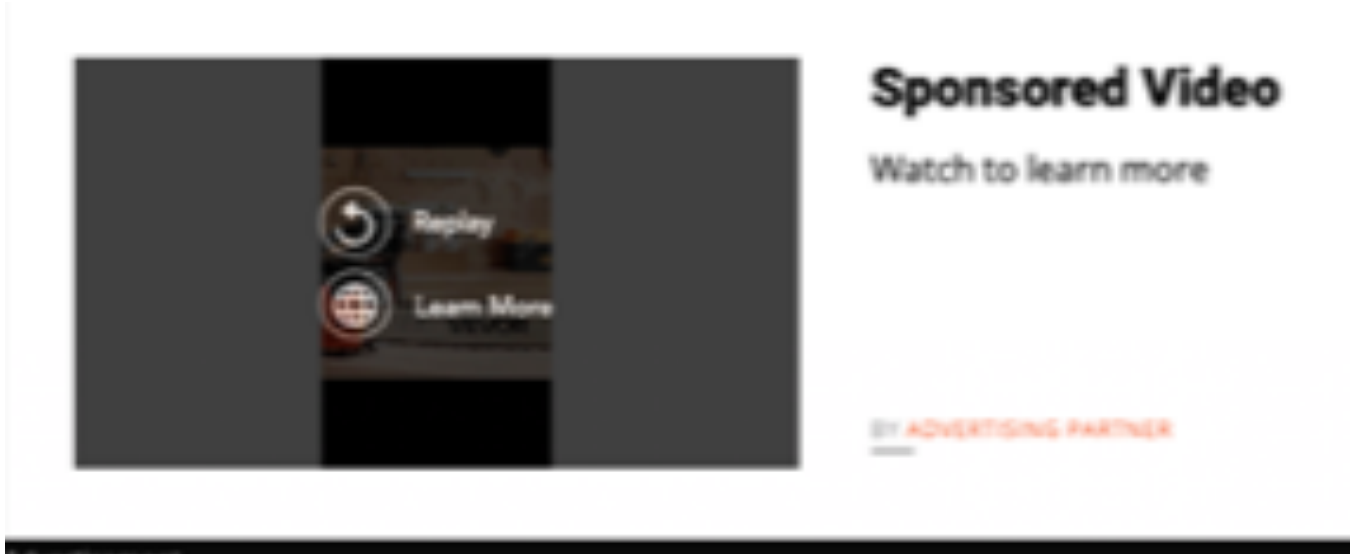
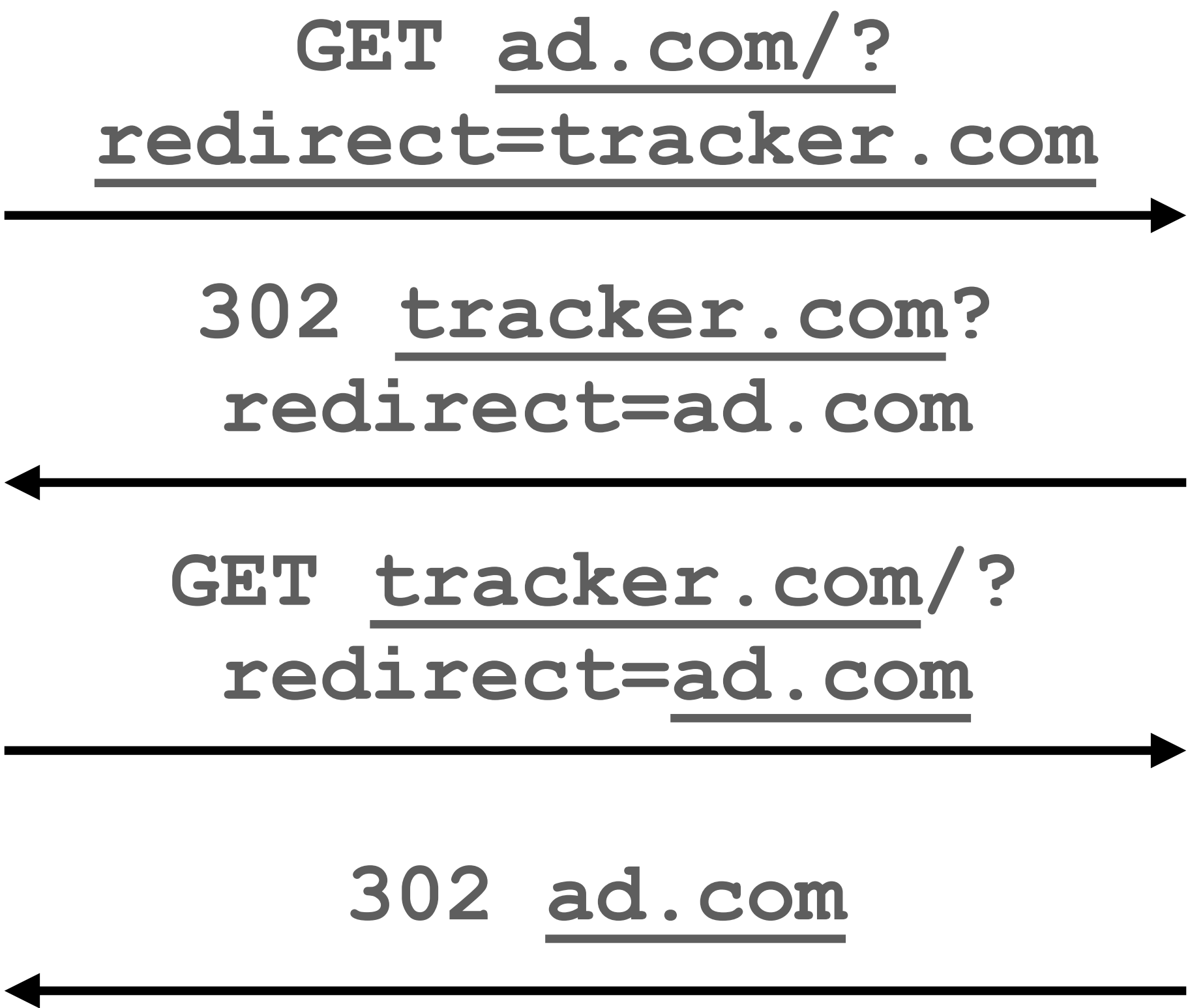
Client



Bounce Tracking



Client



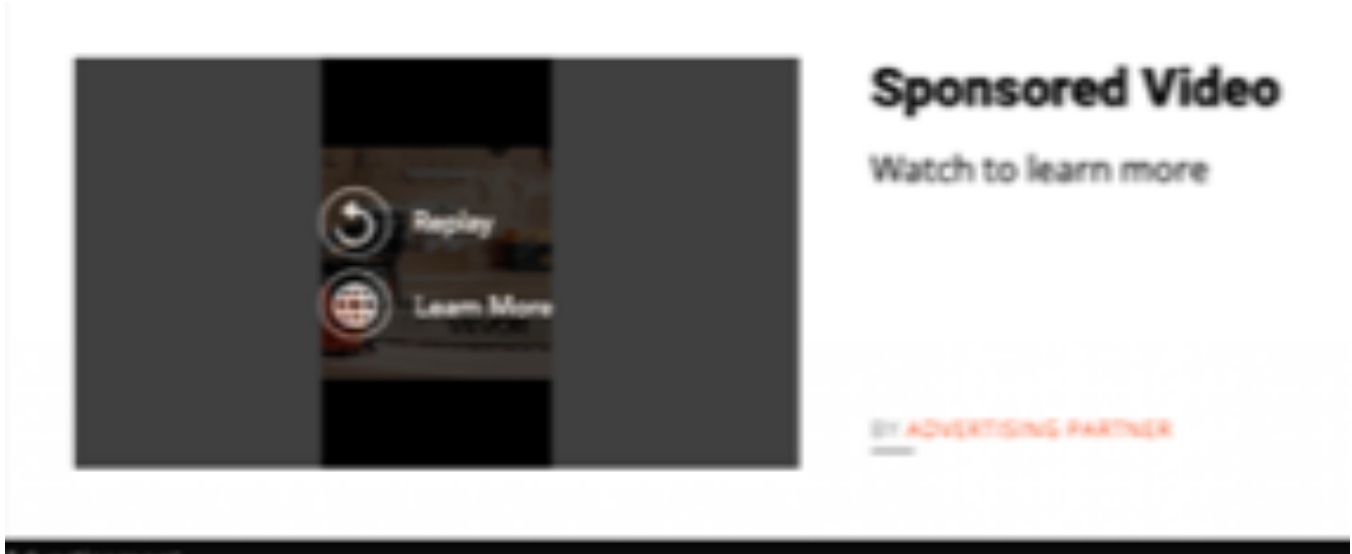
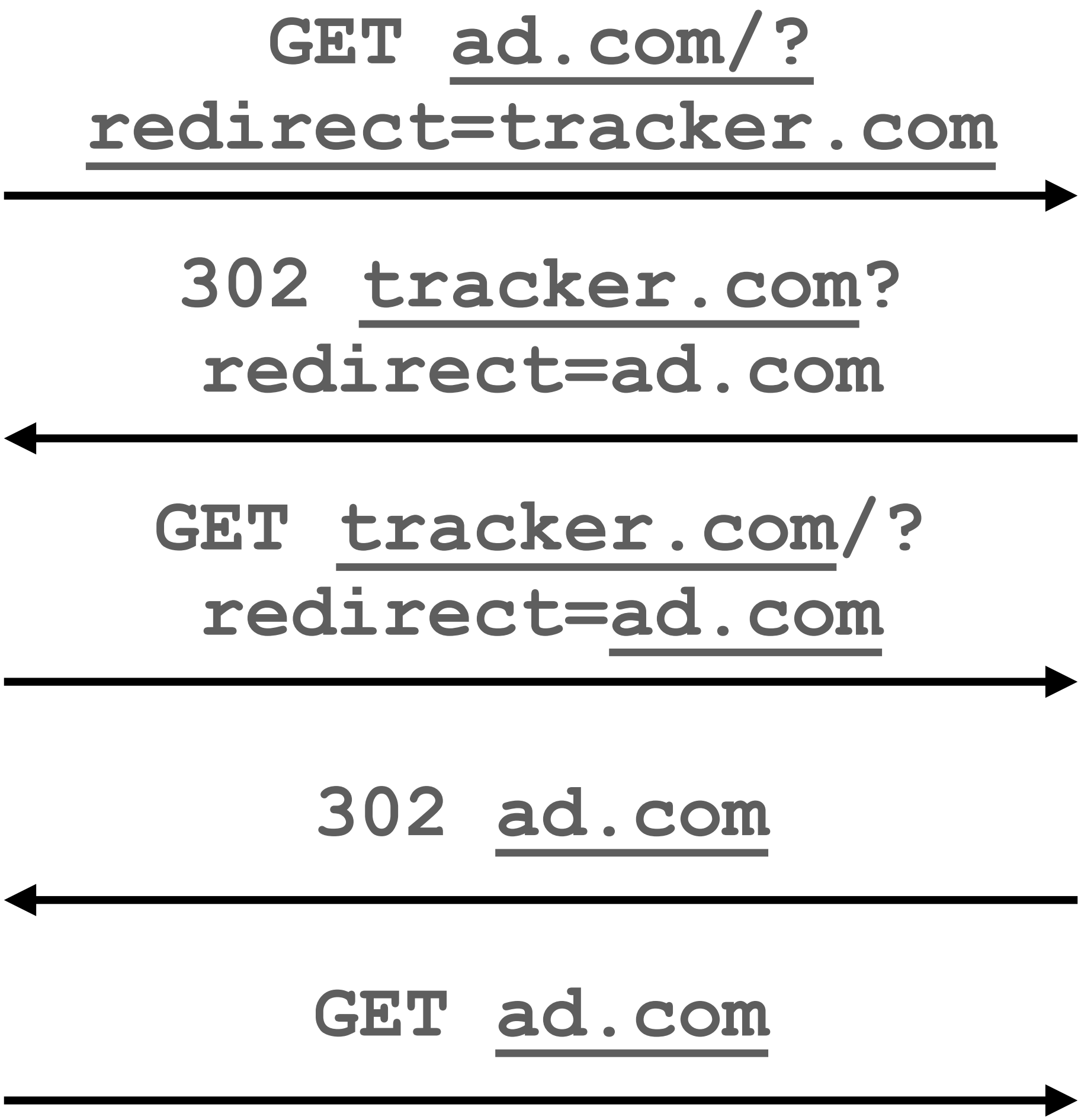
tracker.com

ad.com

Bounce Tracking



Client



Bounce Tracking



Bounce Tracking



Bounce Tracking



How does Bounce Tracking Bypass Third-Party Cookie Protections?

- Bounce Tracking *briefly* redirects your **entire page** to a different tracker, briefly giving them first-party access to your browser (and therefore, setting a cookie)
- Since browsers don't stop first-party requests, this remains totally possible today on most browsers
- Hard to defend against, since it's a "choice" — you could block redirects, but you might break the web
- But, OK, the trackers still have separate identifiers, right?

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- But, OK, the trackers still have separate identifiers, right?
 - **WRONG!**

UID Smuggling

- How can we smuggle identifiers from partitioned cookie stores?



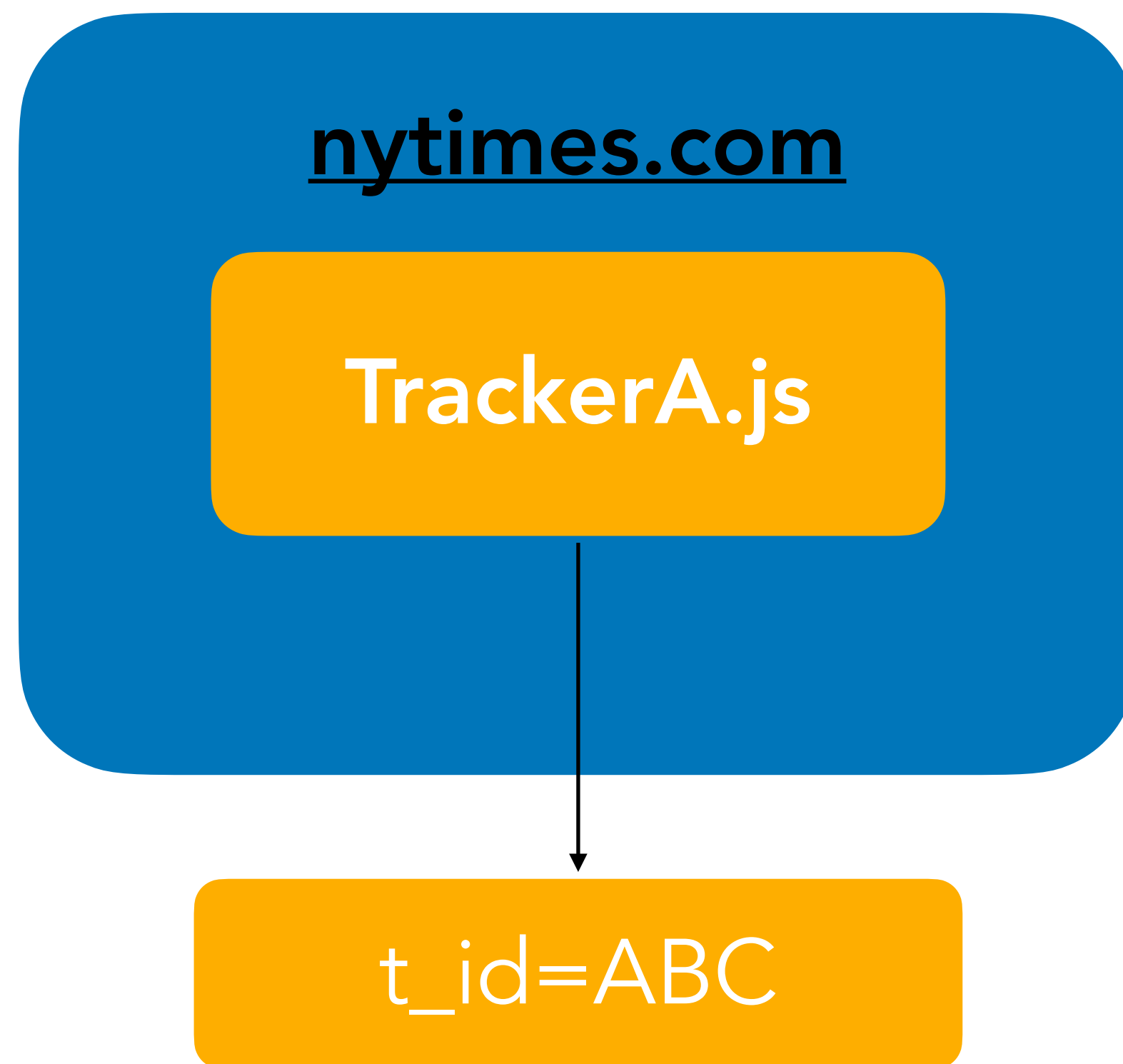
UID Smuggling

- TrackerA.js is included on NYTimes page. This means TrackerA.js can set first-party cookies.



UID Smuggling

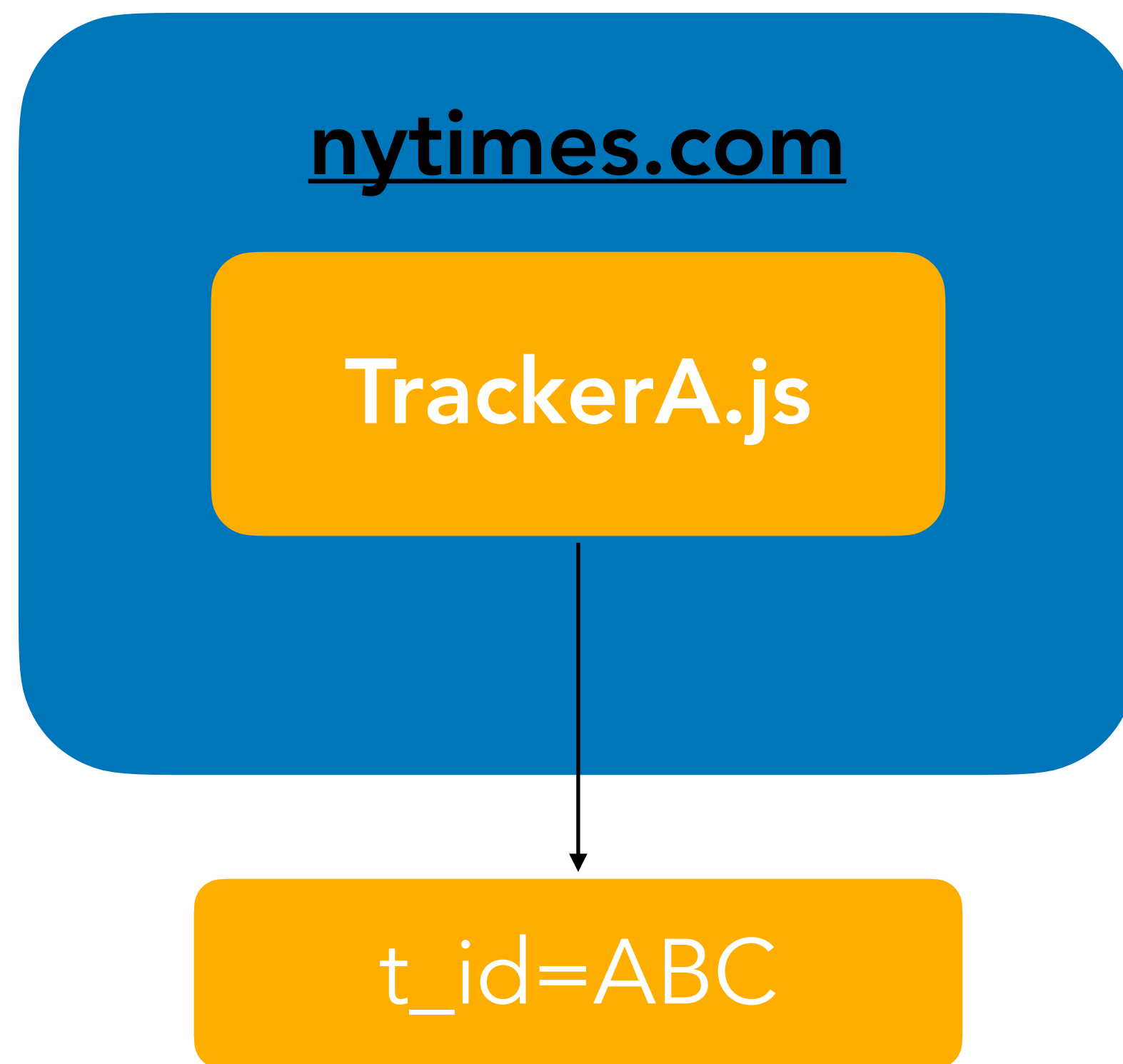
- TrackerA.js is included on NYTimes page. This means TrackerA.js can set first-party cookies.



What is the domain scope of this cookie set to?

UID Smuggling

- TrackerA.js is included on NYTimes page. This means TrackerA.js can set first-party cookies.



What is the domain scope of this cookie set to? nytimes.com + trackera.com

UID Smuggling

- Let's say there's a link to cnn.com on the nytimes.com, and TrackerA knows the JS also runs in cnn.com



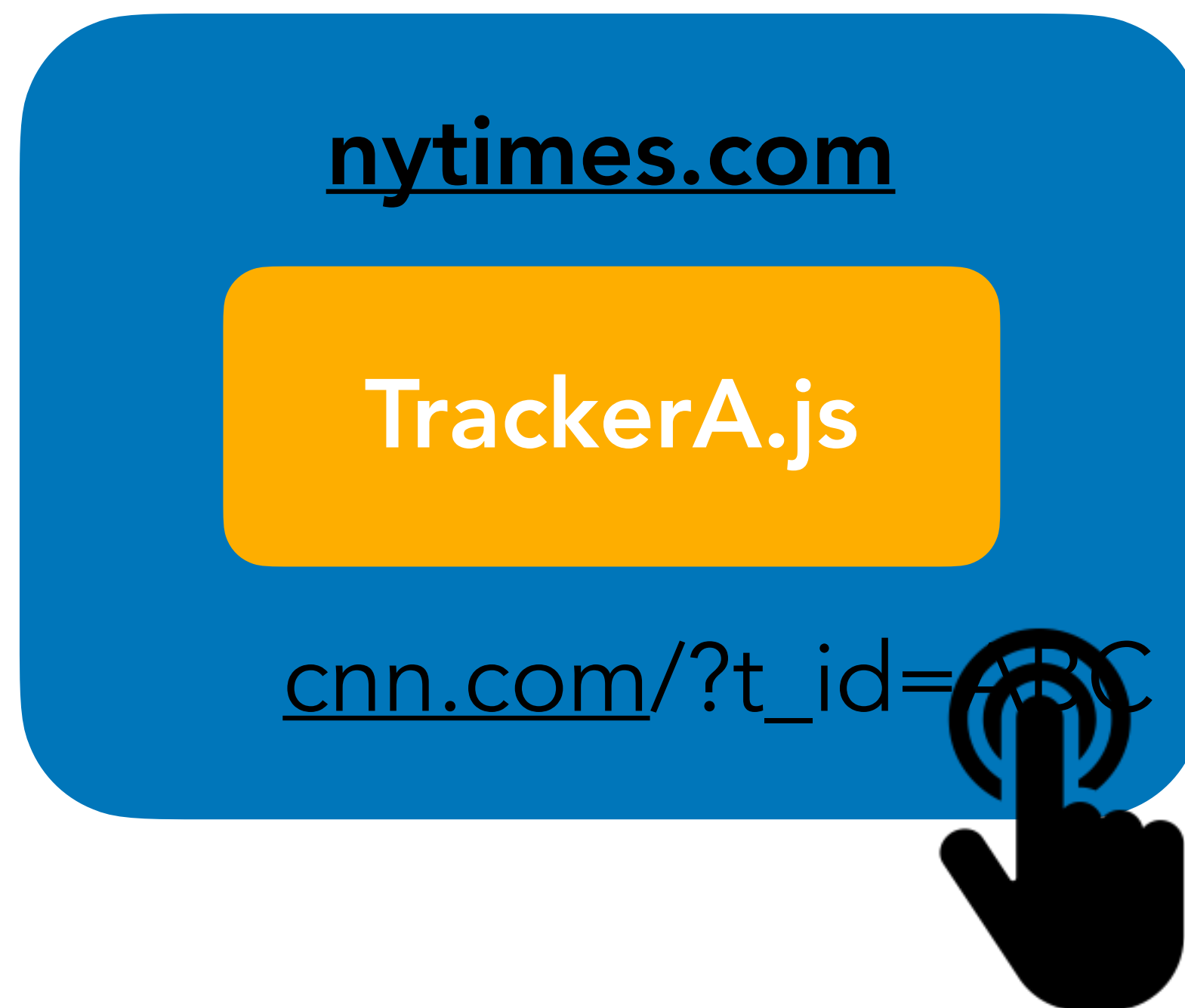
UID Smuggling

- TrackerA is going to rewrite the link to cnn.com to *smuggle* a UID.



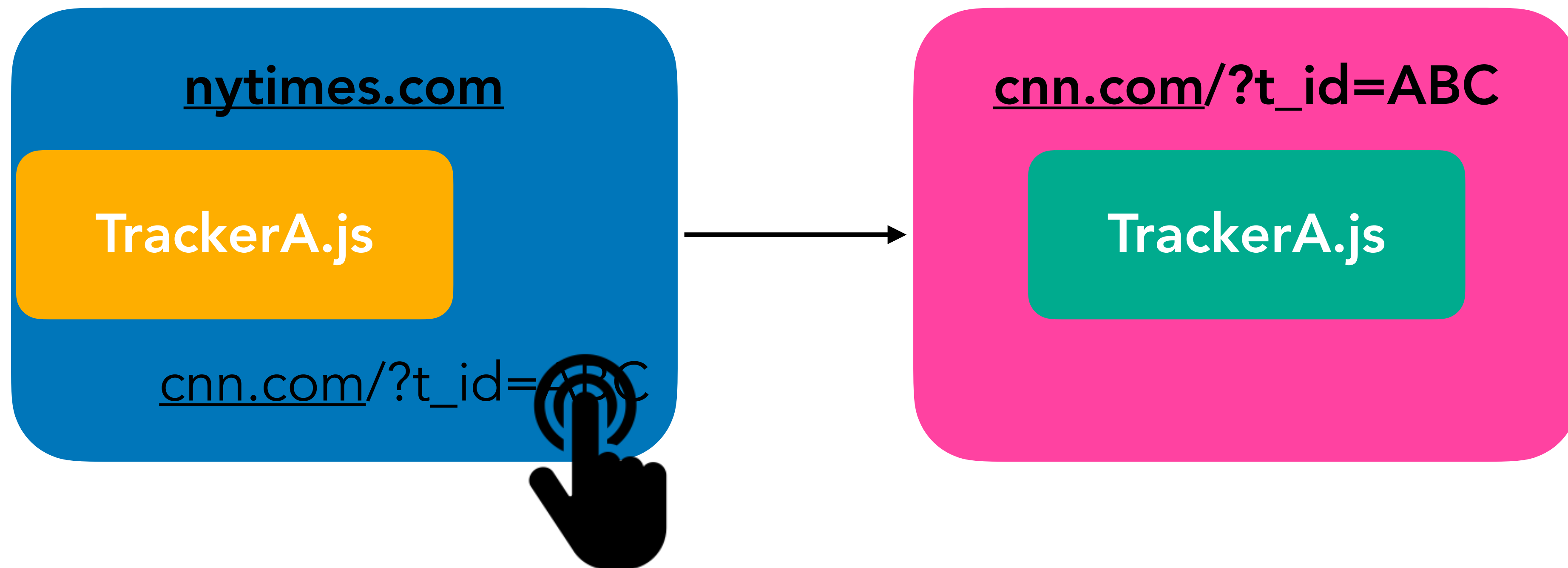
UID Smuggling

- When a user clicks on the link to cnn.com...



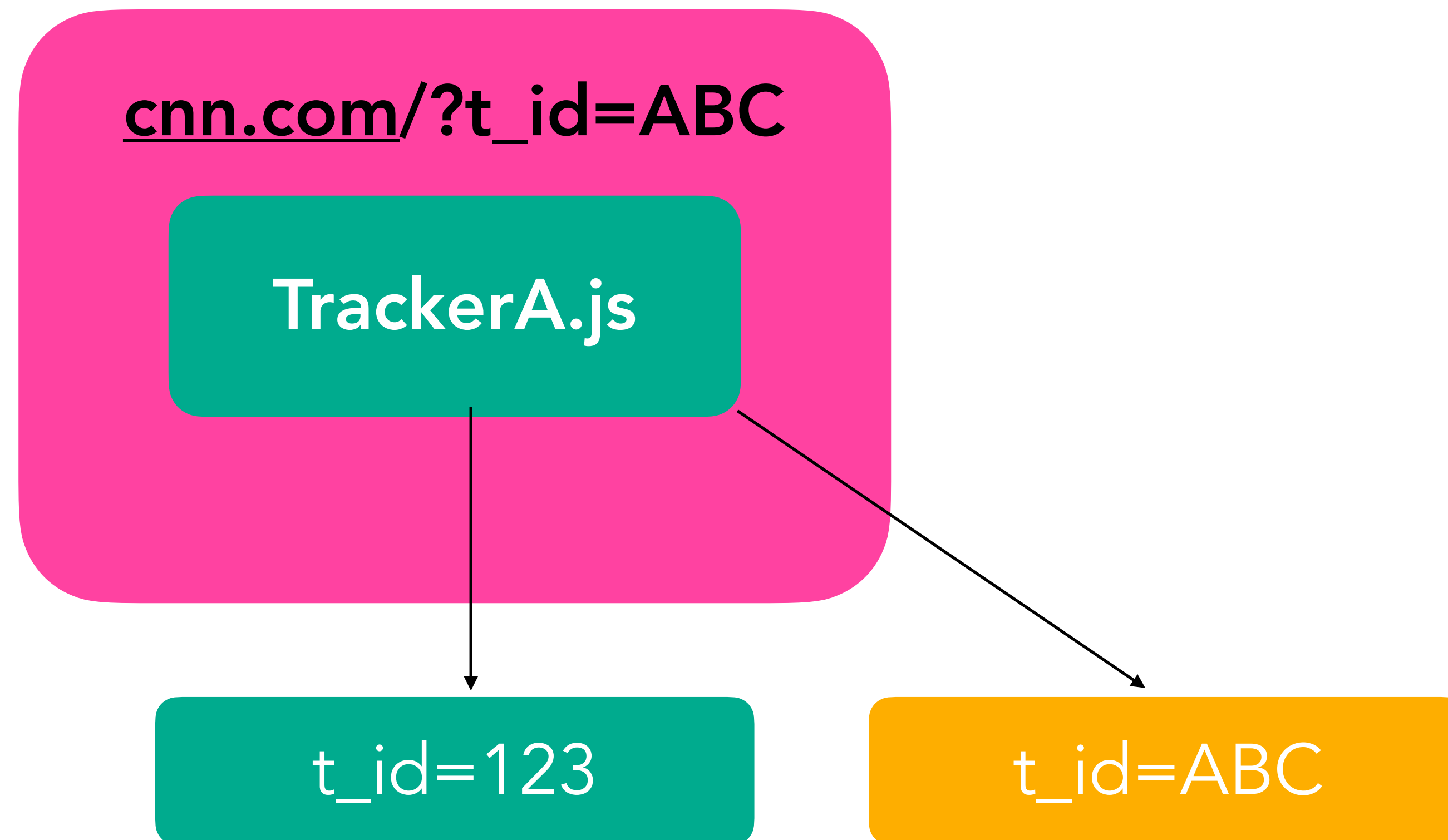
UID Smuggling

- When a user clicks on the link to `cnn.com...` they are routed directly to that page *with* the identifier in the URL



UID Smuggling

- TrackerA has its own cookie for cnn.com... but also, can read the UID from the URL bar! Now trackerA knows these identifiers are the same, and can link in a backend DB



Defeating UID smuggling

- Any ideas?

Defeating UID smuggling

- Any ideas?
- **Remove all query parameters from browser navigation.** Is this a good idea?
 - Might break the web. Might not. Could be interesting research!

Comparing Tracking Methods

	Cookie Syncing	Bounce Tracking	UID Smuggling
Browser mechanism	Tracker-to-Tracker redirect with syncURL	Routing users through redirects	Tracker appends ID to navigation URL
User Action?	Nothing!	User has to click on a link	User has to click on a link
Defense	Partition, Block Third-Party Cookies	Challenging to defend against	Remove all query parameters

Measuring UID Smuggling in the Wild

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UC San Diego

Small aside: Pete from Brave will be giving a guest lecture on 11/3!

Other Strategies: CNAME Cloaking

- A CNAME is basically a pointer to another domain, tells the DNS hierarchy to “look over here” for the IP
- Basic idea — embed yourself **as a first-party** on the webpage by having a DNS record entry point to your domain
 - Bypasses third-party protection altogether

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```
11:16:51 kumarde@macbook ~ dig smetrics.cnn.com

; <<>> DiG 9.10.6 <<>> smetrics.cnn.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24075
;; flags: qr rd ra; QUERY: 1, ANSWER: 11, AUTHORITY: 8, ADDITIONAL: 17

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;smetrics.cnn.com.                IN      A

;; ANSWER SECTION:
smetrics.cnn.com.                300     IN      CNAME   cnn.com.102.122.2o7.net.
```

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;smetrics.cnn.com.                IN      A

;; ANSWER SECTION:
smetrics.cnn.com.                300     IN      CNAME   cnn.com.102.122.2o7.net.
```

**2o7.net is owned by Adobe Analytics...
meaning CNN cookies are sent to Adobe
by the browser by default :)**

What can you do about it?

- Ultimately, your choice... but an easy way to control your privacy is to use a browser that values your privacy
 - Firefox, Safari, Brave, Tor (if you wish)... all of these do much better than Chrome by default
- Install ad blockers and privacy extensions!
 - UBlock Origin, Ghostery, Privacy Badger, Global Privacy Control... lots of ways to have a little bit more control over your data
- Stay informed!
 - Mozilla, Safari, Chrome, etc. all have blogs that regularly write about this stuff

Next time...

- **NO CLASS on 10/13!** Work on your PAs!
- Next Thursday, Kristen will be discussing inference... the basics of how companies use this data they collect about you to *actually infer things about you!*
- PA1 parts 3 + 4 will release next Wednesday!